McKinsey & Company

Chilean Hydrogen Pathway

Final Report

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Chapter 1: State of global Hydrogen industry

Chapter content description

- **State of the global Hydrogen industry, including**: production, transportation, distribution and utilization of Hydrogen (specific focus on Green Hydrogen)
- **Development of hydrogen market, including:** hydrogen projects and investment analysis by key market and segment of the hydrogen value chain
- **Roadmaps and strategies for hydrogen development in key markets** including key strategic initiatives, incentive mechanisms and other forms of public support

Activities included

- **Activity 1.1**
- **Activity 1.2**

1.1/ Blue and Green hydrogen are two low-carbon production methods of H2 with the highest future potential

X **CO2 emissions**
kg CO2 per kgH2 produced

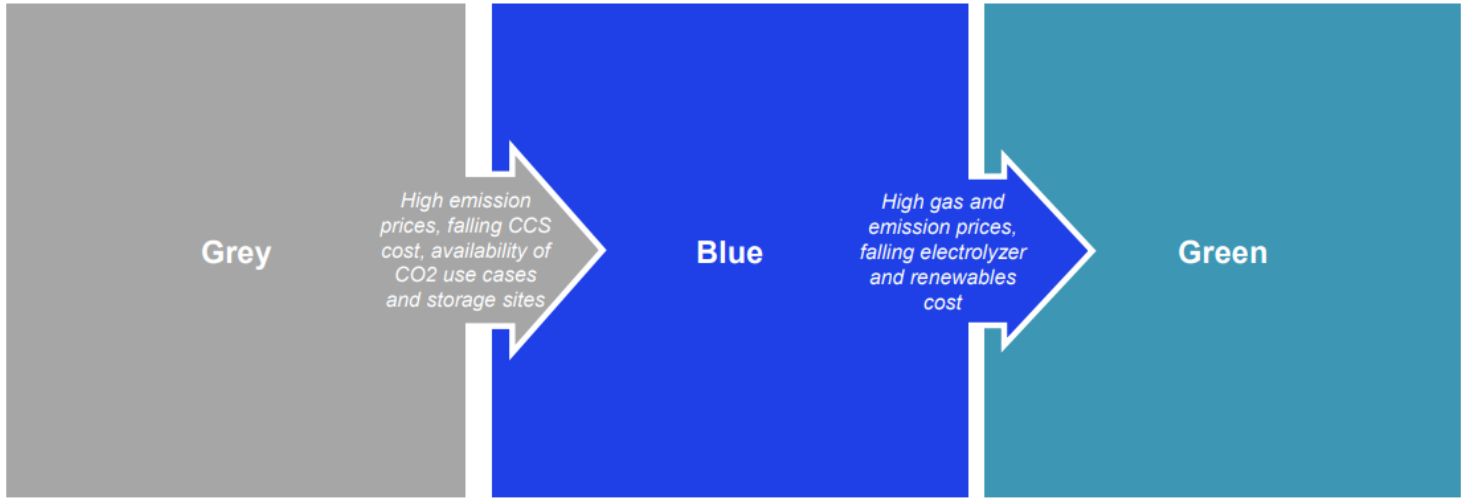
	Grey H2	*Low-carbon H2* Blue H2	Green H2
Production process	Split[1] natural gas into H2 and CO2 ~10	Similar to Grey but additionally capture, potentially use, and store CO2 ~1-3 *Most CO2 stored*	Split water into H2 and O2 in an electrolyzer that is powered by renewables ~0 *Assuming Green electricity mix[2]*
Investments and complexity	Large scale, one-off facilities	Large scale and one-off facilities	Small-scale (5-10MW) facilities that can be replicated, similar to utility-scale batteries[3]
	Requiring triple-digit million EURCAPEX per facility	Requiring triple-digit mn EUR CAPEX per facility	Requiring single/double-digit EUR mn CAPEX per facility, expecting 50-70% reduction until 2030
	3-5 years EPC	5-10+ years EPC duration, contingent upon development of CO2 storage site	1-3 years EPC duration
Direct link to power sector	✗	✗	✓

1. Process: Sulfur Removal, Synthesis Gas Production via Steam-Methane Reforming (SMR) or Auto-Thermal Reforming (ATR), CO Shift Reaction, Purification. The latter is expected to offer higher efficiencies in combination with CCS. In addition, Grey hydrogen can also be produced from coal gasfication.
2. In Australia, producing hydrogen from electrolyzers that run on grid electricity would lead to an emission intensity of ~40 kgCO2/kgH2.
3. Especially PEM electrolyzers can be largely pre-assembled, containerized, and stacked.

1.1/ Blue H2 is cheaper today, but Green H2 is the ´new solar´ with costs decreasing rapidly

Global H2 demand and production costs[1]

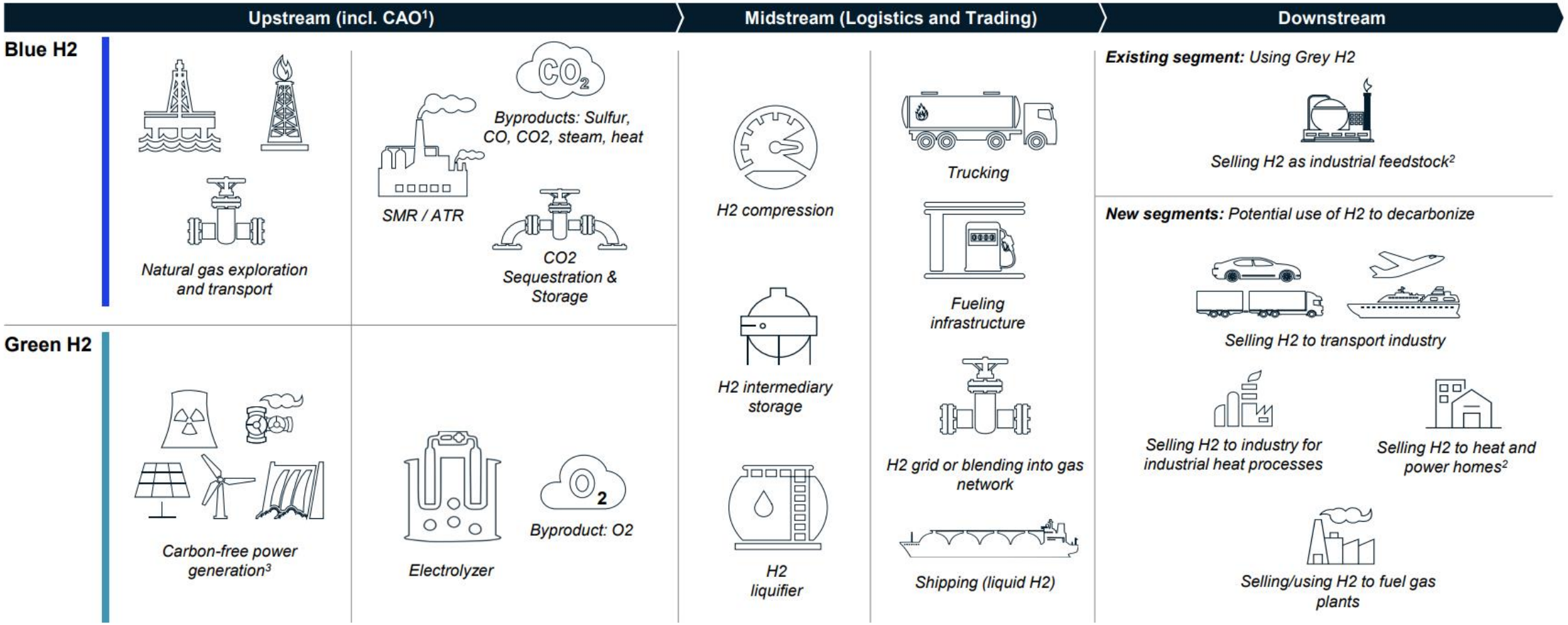
■ Least-cost option □ 2nd least-cost option



	Global H2 demand Mt p.a.	Thereof Grey H2 Mt p.a. (Share %)	Thereof Blue+Green H2 Mt p.a.	Production cost Grey H2 $/kg	Blue H2 $/kg	Green H2 $/kg
2018	73	73 (100%)	~0	■ ~1.0-2.5	□ ~1.5-3.0	4.3
2030	110	73+X (>70%)	Up to 37	■ ~1.0-2.5	□ ~1.5-3.0	□ 2.0
2050	545	73+/-X (<20%)	Up to 472	"	"	■ 1.3

1. Estimates by Hydrogen Council; Grey H2 production costs assuming natural gas prices of $4-$8/mmBtu; Blue H2 costs assuming Grey costs + $0.5/kg; Green H2 costs assuming IRENA database weighted average solar PV costs (auctions and PPAs) in 2020

1.1/ The Green H2 production relies on carbon-free power generation and electrolyzers which split water H2 and an O2 biproduct



1. Commercial Asset Optimization
2. Refining, ammonia, methanol, olefin, steel 2 Blending H2 in natural gas networks or dedicated H2 networks 3 Water electrolysis based on "grey" electricity mixes may result in "fake-green" H2 with higher emissions than Grey H2

1.1 / Hydrogen's value chain is decomposed into production ('green' or not), handling and final use (across several verticals)

Production (techniques)[1]

Reforming/ gasification of fossil feedstock

100%+ of current production. Natural gas, oil, coal or biogas is used to get hydrogen:

- Steam methane reforming (SMR)
- Petroliquids or coal gasification

Can be associated to a carbon capture & storage facility to make H_2 production more sustainable

Electrolysis

Splitting of water to get hydrogen, using renewable or fossil fuel-based energy.

- 2 power sources: on-site or from the grid
- 3 main technologies: alkaline (most mature), PEM[1] (increasing interest) and high-temperature splitting (nascent)

Other techniques

Other (nascent) techniques to produce hydrogen include:

- Biomass gasification
- Biological / Bacterial production
- Direct solar water splitting

Handling (steps)

Storage

Storage of hydrogen or hydrogen-based fuels & feedstock

- For short duration & small volumes, storing in tanks and / or solid materials
- For longer duration & large volumes, storing in salt caverns, depleted gas wells or aquifers

Transmission

Transmission of hydrogen for long distances / large flows

- Can involve pipelines (H_2 or methane), rail transport, ships and / or trucks
- Can transport hydrogen in a gaseous or liquid form, or transport liquid organic H_2 carriers

Distribution

Transmission of hydrogen for shorter distances to the final user

- Can involve pipelines and / or trucks
- Can transport hydrogen in a gaseous or liquid form, or transport liquid organic H_2 carriers

Final use

Power storage & generation

- Battery storage / Buffer
- Power generation

Transport

- Cars
- Vans / Minibuses
- Trucks
- Ships & Planes
- Forklifts
- Trains & Trams

Industry feedstock

- Oil refining
- Ammonia / Methanol production
- Steel (iron ore reduction)

Industrial energy

- Low industry heat
- Medium industry heat
- High industry heat

Building heat and power

- Heating
- Onsite power generation

1. Hydrogen produced may then be transformed to hydrogen carriers (ammonia, methanol, etc.)
2. Polymer electrolyte membrane

Source: IEA 2019; McKinsey analysis

1.1 / Since 2018, the hydrogen industry has been experiencing unprecedented momentum

Not Exhaustive

Transportation · Industrial energy · Building heat · Power generation/storage · Overarching · Hydrogen Production · Hydrogen Infrastructure

2018

- Jan 18 — Chinese plans to build **hydrogen infrastructure** for ~50k FCEVs by 2025
- Mar 5 — Foundation of **Japan H2 Mobility**
- Mar 29 — **Shell and ITM Power** agree to build then **world's largest electrolysis plant (10 MW)** in Germany
- May 2 — **Toyota** aims for 30k FCEV sales per year after 2020 – build **two new fuel cell facilities**
- May 14 — **Hydrogen roadmap for the Netherlands**
- May 24 — **French strategy plan:** France as a **leader in hydrogen technology**
- Jun 4 — **FCEV taxi fleet** in Paris expanded to 100 FCEVs
- Jun 11 — Announcement of a test flight of ZeroAvia hydrogen-powered six-seater airplane
- Jul 25 — **Engie's power-to-gas demo project** starts injection of **6% hydrogen** in the gas grid, rising to 20%
- Aug 14 — Chinese **Weichai** strategically invests USD 163m in **Ballard**
- Aug 15 — **Siemens** and **PowerCell** announce development of **fuel cells for ships**
- Aug 22 — **NEL** announces to build **world's largest electrolyzer production plant** (360 MW/year) in early 2020
- Aug 29 — **EU energy ministers** announce joint hydrogen R&D
- Sep 5 — **First hydrogen train** starts operating in Germany
- Sep 17 — **Hydrogen Council** expands to **53 members**
- Sep 18 — **VW and Stanford University** announces significant reduction in platinum use for fuel cells
- Sep 27 — **New "H2Bus Europe" funding program** for 600 fuel cell buses (EUR 40m)
- Sep 28 — **ITM Power** gets funding to **explore 100 MW power-to-gas storage project**
- Oct 3 — **Hyundai** announces supply of **1,000 hydrogen fuel cell lorries** to Swiss H2 Energy
- Oct 10 — **Hyundai** announces to sell **5,000 FCEVs to France** by 2025
- Oct 12 — **Northeast** of the US announces start of HRS roll-out
- Oct 14 — **Australian Renewable Energy Agency** announces **2-year trial** injecting green hydrogen into **Sydney's gas grid**
- Oct 16 — **100 MW power-to-gas project** by Gasunie, Tennet and Thyssengas starting 2022
- Oct 22 — AirLiquide and Linde announce expansion of liquid H2 production in US
- Nov — Hyundai announces FCEV Vision 2030: 7bn USD invest, 700.000 units by 2030
- Dec 12 — China beats FCEV target of 2020 in 2018, on track to become largest fuel cell market

2019

- Jan 25 — Yara and Engie to build 20 MW green hydrogen plant
- Feb 5 — tk announces carbon-neutrality by 2050, $10bn EUR investments
- Feb 18 — Orsted includes H2 in Dutch offshore auction
- Mar — AirLiquide acquires ~20% of Hydrogenics
- Apr 8 — EDF launches subsidiary to develop hydrogen solutions for industry
- June 15 — IEA launches hydrogen report with major reversal of their perspective, report endorsed by G20 energy ministers
- June 19 — Cummins acquires Hydrogenics
- June 29 — RWE and Innogy announce plans for 100 MW electrolyzer in Netherlands
- Aug 19 — Ship-maker CMB announces several hydrogen investments
- Aug 27 — **Linde** announces quadrupling of hydrogen and syngas capacity at $1.9b Jurong Island complex
- Sep 3 — Bosch and Hanwa invest over $230m in Nikola
- Sep 5 — CNH announces $250m investment in Nikola and JV for hydrogen truck production

Key trends in 2019

Acceleration of momentum as **decarbonization commitments become more important**

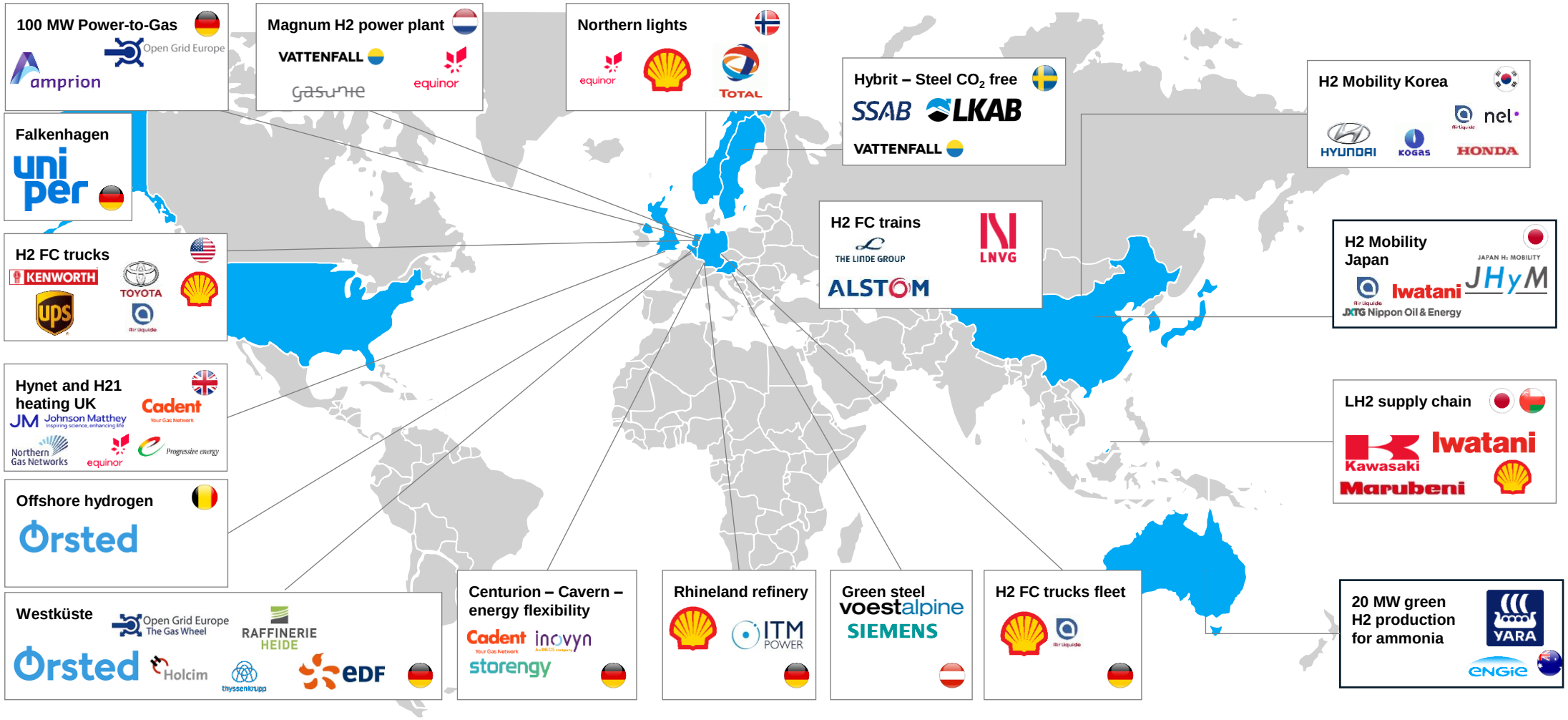
New segments growing, e.g. trucks, trains, ships, synfuel

Push from utilities, renewable developers and oil & gas companies

Large-scale projects with water electrolysis, mostly for use in industry (fertilizers, refining)

Source: Press releases; McKinsey

1.1 / Investment has been led by large industrials teaming up for large flagship projects – to reach technological and commercial maturity



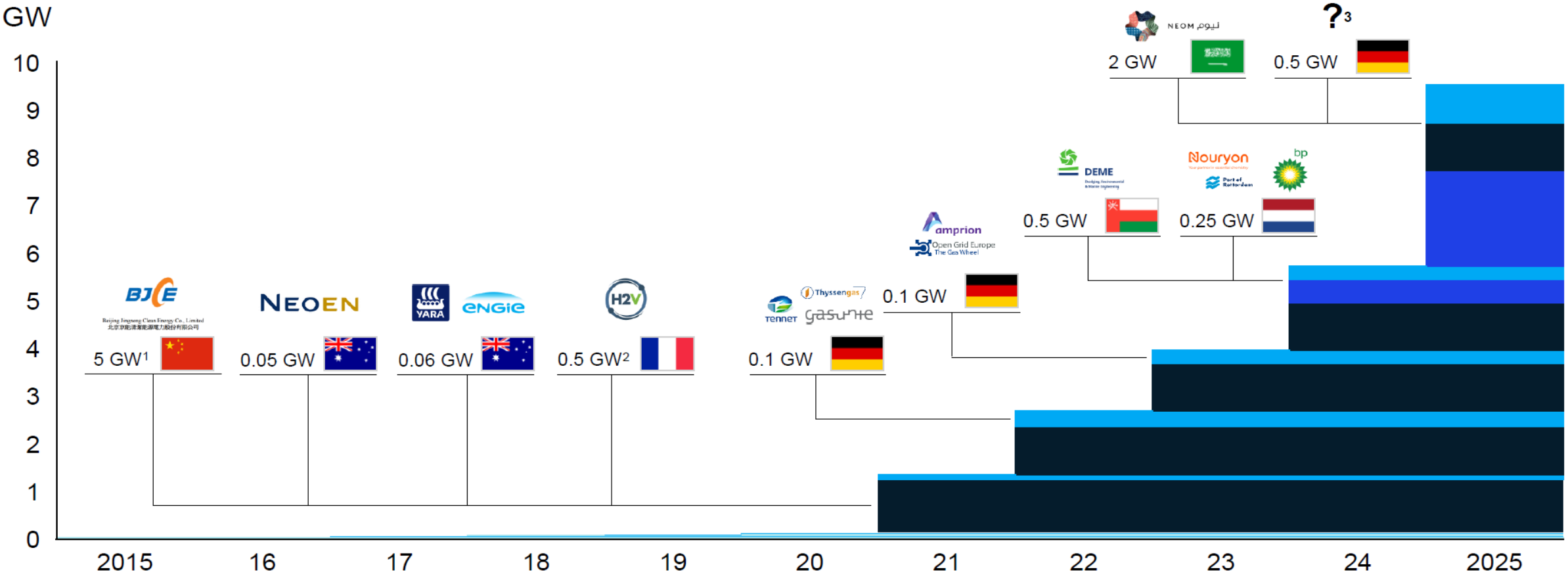
Source: Press releases; McKinsey

1.1 / Global green H2 production capacities are limited, but expected to increase significantly with ~9.5 GW announced until 2025

Selected project examples

Asia Pacific | Europe | Middle East | North America | Africa | Latin America

Global electrolysis project announcements,
GW



1. Plant assumed to be constructed over 5 years 2. Various plants announced to be constructed until 2025 3. Government announcements, most likely allocated through tenders/auction

Source: IEA, Press research, McKinsey

1.2 / Countries have establish hydrogen roadmaps with different levels of detail on key issues for the promotion of the industry

Detailed next | ✓ Mentioned but not detailed | ✓ Considered in detail | PRELIMINARY | NOT EXHAUSTIVE

		European Union	Netherlands	Germany	South Korea	Australia	Chile
Indicators	H2 production projection (Mton)	1Mton by 2024 10Mton to 2040	NA	NA	0.5 Mton to 2022 5 Mton to 2040	NA	11M 2040
Indicators	Installed electrolysis Capacity	6GW by 2024 40GW by 2030	0.5 GW by 2025 3-4GW by 2030	5GW by 2030 10GW by 2035/40	NA	NA	5 GW 2025 25GW 2030
Indicators	Investment	180-470 EUR B by 2050	NA	9 EUR B to boost h2 demand	2.3 USD B to 2022 for the H2 vehicle industry	NA	45 USD b 2030 +300 USD b 2050
Domains	1 Vision and targets	✓	✓	✓	✓	(✓)	✓
Domains	2 Regulation and licensing	✓	✓	(✓)	✓	(✓)	✓
Domains	3 Coordination and partnerships	✓	✓	✓	✓	✓	✓
Domains	4 Financing and incentives	(✓)	✓	✓	✓	✓	✓
Domains	5 Infrastructure	✓	✓	✓	✓	✓	(✓)
Domains	6 Research & Development	✓	✓	✓	✓	✓	✓
Domains	7 Skilled Labor	(✓)		(✓)	✓	(✓)	(✓)

1.2 / The EU Hydrogen Strategy For a Climate Neutral Europe

NOT EXHAUSTIVE

General approach

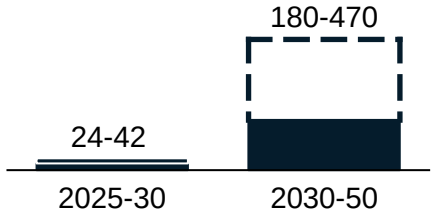
The EU has agreed on its **EU Hydrogen strategy** as a **roadmap** to build up a large-scale **green hydrogen industry**

The roadmap is divided into **3 phases** until **2050** that aim at reaching **increasing levels** of **maturity for the H_2 market**

The strategy explores how green hydrogen can help reducing the **EU economy's carbon emissions** in a **cost-effective way**

It is in line with the **EU's goal** to be **climate-neutral by 2050**

Cumulative investment needs in Europe, EUR bn



Overview of key action areas

Vision & targets	Build a concrete **project pipeline** by end of 2020 Develop an **investment agenda and** support further **strategic investments**
Coordination and partnership	Strengthen **EU leadership** internationally Start cooperation with **Southern and Eastern Neighborhood** partners, **Energy Community countries, and** the **African Union**
Financing and Incentives	Explore **support policies** to **increase demand** Implement **EU-wide criteria for low-carbon H_2 certification** and introduce a **common threshold** Develop a pilot scheme for a **Carbon CfD[2] program**
Infrastructure	Build up **infrastructure** for **fueling and transport** Design **market rules** to ensure **liquid markets** and to **remove barriers** for **repurposing gas infrastructure**
R&D	Develop **pilot projects** that test full **value chains** Facilitate **demonstration of H_2-based technologies**

Key commitments and targets

6 GW Of installed electrolysis capacity by 2024

40 GW Of installed electrolysis capacity by 2030

Green hydrogen production targets, Million tons

11. Contracts for Difference

Source: European Commission 2020 - A hydrogen strategy for a climate-neutral Europe

1.2 / The EU's Hydrogen Strategy aims at achieving a renewable hydrogen mass market based on a 3-phase-roadmap

x Related amount of green hydrogen production in the EU, million tons

	Phase 1: 2020-2024	Phase 2: 2025-2030	Phase 3: 2030-2050
Objective	**Market launch** – Decarbonize existing hydrogen production and facilitating take up of hydrogen consumption in new end-uses	**Scale up** – Expand production and end-use to additional sectors to develop hydrogen as intrinsic part of an integrated energy system	**Mass market** – Deploy renewable hydrogen at large scale to reach all relevant hard-to-decarbonize sectors
Electrolysis capacity targets, GW	1 6	10 40	? [redacted]
Infrastructure roadmap	Mostly **local on-site supply** combined with selective **gas blending** and build-out start of comprehensive **refueling infrastructure**[1] Planning of **medium range and backbone transmission** infrastructure	Expansion to **regional hydrogen clusters** with **transport over short distances** also through **dedicated hydrogen pipelines** Need for an **EU-wide logistical infrastructure** to emerge 400 additional **refueling stations** (no year specified)	Expansion to **EU-wide hydrogen infrastructure** and **hydrogen trading with non-EU** partners
Targeted end-use sectors[2]	**Transport** (buses, trucks) **Heavy industries** (refineries, steel, chemicals)	**Transport** (rail, selected maritime, others) **Heating** (residential, commercial) **Power system flexibility** (storage, backup, buffering)	**Transport** (aviation, long distance maritime) **Synthetic fuels production**
Selected support schemes	**Revision** of the **ETS** and potentially a **carbon border adjustment** mechanism **Carbon contracts for difference**[3], competitive **tenders and quotas** for end-use sectors The upcoming **Sustainable and Smart Mobility Strategy** adds further support for **transport uses**		**No dedicated support schemes** stated for Phase 3 as hydrogen mass market **expected to carry itself**

1. Refueling infrastructure build-out to continue across phases in line with increase in demand for transport applications 2. From phase 2 targeted end-use sectors refers to additions to the sectors mentioned in the previous phase(s)
3. Scheme potentially to be applied within replacement of existing hydrogen production in refineries and fertilizer production, low carbon and circular steel or maritime and aviation applications 4. Focus on hydrogen production capacities

Source: European Commission 2020 - A hydrogen strategy for a climate-neutral Europe

1.2 / Key actions in the EU's Hydrogen Strategy for a climate-neutral Europe – Enabling environment

NOT EXHAUSTIVE

Domain	Key measures	Description
Vision and targets	**Introducing a roadmap to 2050 for a hydrogen ecosystem**	Establish a clear vision **on hydrogen production targets and deployment timeline,** as well the most relevant end-use sectors per phase Introduce actions around **5 key dimensions**: develop an investment agenda, boost demand an scale up production, design an enabling framework, promote research and innovation, strengthen the international dimension Provide a broad forum to coordinate investment by all stakeholders and engage civil society, through sector-based round tables and a policy-makers' platform, supervised by the **European Clean Hydrogen Alliance**
Regulation and licensing	**Designing and enabling market rules**	Design enabling **market rules for the deployment of hydrogen**, including removing barriers for efficient hydrogen infrastructure development (e.g. via repurposing), and ensure access to **liquid markets for hydrogen producers and customers** and the integrity of the internal gas market, through the upcoming legislative reviews (e.g. review of the gas legislation for competitive decarbonized gas markets) (2021) Develop third-party access rules, **clear rules** on connecting electrolyzers to the grid and **streamlining of permitting** and administrative hurdles to reduce undue burden to market access Work to introduce **a comprehensive terminology and European-wide criteria for the certification** of renewable and low-carbon hydrogen (by June 2021) Propose measures to facilitate the use of hydrogen and its derivatives in the transport sector in the Commission's upcoming **Sustainable and Smart Mobility Strategy**, and in related policy initiatives (2020)
Coordination and partnerships	**Reinforcing EU leadership in technical standards and regulations**	**Strengthen EU leadership in international technical standards, regulations and definitions** on hydrogen **Develop the hydrogen mission** within the next mandate of Mission Innovation (MI2) Promote cooperation with **Southern and Eastern Neighborhood partners and Energy Community countries, notably Ukraine** on renewable electricity and hydrogen Set out a **cooperation process on renewable hydrogen with the African Union** in the framework of the Africa-Europe Green Energy Initiative Develop a **benchmark for euro denominated transactions** by 2021

Source: European Commission 2020 - A hydrogen strategy for a climate-neutral Europe

1.2 / Key actions in the EU's Hydrogen Strategy for a climate-neutral Europe – Investments and incentives

NOT EXHAUSTIVE

Domain	Key measures	Description
Financing and incentives	**Developing an investment agenda for the EU**	Through the European Clean Hydrogen Alliance, **develop an investment agenda** to stimulate the roll out of production and use of hydrogen and **build a concrete pipeline of projects** (by end of 2020) Support strategic investments in clean hydrogen in the context of the **Commission's recovery plan**, in particular through the Strategic European Investment Window of **InvestEU** (from 2021) Explore additional support measures, including **demand-side policies in end-use sectors and a carbon Contracts for Difference** program
Infrastructure	**Designing a framework for infrastructure**	**Start the planning of hydrogen infrastructure**, including in the Trans-European Networks for Energy and Transport and the Ten-Year Network Development Plans (TYNDPs) (2021) taking into account also the planning of a network of refueling stations Accelerate the **deployment of different refueling infrastructure** in the revision of the Alternative Fuels Infrastructure Directive and the revision of the Regulation on the Trans-European Transport Network (2021)
R&D	**Promoting research and innovation in hydrogen technologies**	**Launch a 100 MW electrolyzer and a Green Airports and Ports call for proposals** as part of the European Green Deal call under Horizon 2020 (Q3 2020) Establish the proposed **Clean Hydrogen Partnership**, focusing on renewable hydrogen production, storage, transport, distribution and key components for priority end-uses of clean hydrogen at a competitive price (2021) Steer the development of **key pilot projects that support Hydrogen value chains**, in coordination with the SET Plan (from 2020 onwards) Facilitate the demonstration of innovative hydrogen-based technologies through the launch of calls for proposals under the **ETS Innovation Fund** (first call launched in July 2020) Launch a call for pilot action on **interregional innovation under cohesion policy** on Hydrogen Technologies in carbon-intensive regions (2020)
Skilled labor	**N/A**	Support the needed adjustments in upskilling and in the labor market through the **European Clean Hydrogen Alliance**

Source: European Commission 2020 - A hydrogen strategy for a climate-neutral Europe

1.2 / The EU supports hydrogen innovation, demonstration and scale up through different instruments

Main relevant examples

EU Funding | Financial instruments with risk sharing component | Loans | Technical Assistance

Funding sources		Pre-commercial development		First of a kind		Uptake, market ready, roll-out
		Proof of concept →	Pilot →	Demons-tration →	Scale up →	Roll out
EU	Horizon Europe	→	→	→	→	
	Connecting Europe Facility (CEF)			→	→	→
	LIFE Programme			→	→	→
	Innovation Fund			→	→	→
	InvestEU			→	→	→
	InnovFin		→	→	→	→
	EIB Loans			→	→	→
	EIB Advisory Hub (EIAH)	→	→	→	→	→
Hybrid, blending EU and national funding	European Structural and Investment Funds (ESIF)			→	→	
National Funding on the basis of a European plan	Important Projects of Common Interests (IPCEI)		→	→	→	

1.2 / South Korea Hydrogen Strategy pursues industrial competitiveness and growth

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General approach

Long-term targets until 2040 were released in the **"Hydrogen Economy Roadmap of Korea"** and the **"National Roadmap of Hydrogen Technology Development"** in 2019

Next to **hydrogen mobility**, the strategy focuses on the long-term establishment of a "hydrogen-powered" society, with key areas being the **production of hydrogen vehicles**, hydrogen-related **infrastructure** and hydrogen for **power generation**

Key measures include government funding, subsidies, industry alliances and development partnerships with other countries such as Australia, Saudi Arabia, Norway and Israel

National OEMs and suppliers also announced significant investments into hydrogen, e.g. Hyundai Motor Group and its suppliers plan to invest a total of **USD 6.7 bn** and to produce **500k FCEVs** annually by 2030

Overview of key action areas

Vision and targets	Established targets on **FCEV production and refueling stations** in the country
Regulation and licensing	Establish legal **safety standards** to ensure the reliability and stability of the hydrogen economy
Coordination and partnerships	Establish a **cooperation system among Korea, China, and Japan,** and reinforce participation **in global partnerships** (e.g. Hydrogen Council)
R & D	Create of **a Hydrogen Industry Cluster** (2021) for R&D cooperation between relevant players
Skilled labor	Create a **safety professional qualification** to manage safety across all processes of the hydrogen value chain

Key commitments and targets

No. 1 fuel cell producer worldwide

6.2 Mn FCEV annual production capacity (incl. passenger cars, trucks, and buses) by 2040

2.3 Bn USD for the establishment of a public-private hydrogen vehicle industry ecosystem by 2022

Hydrogen[1] consumption targets,
Million tons per year

1. Includes grey, blue, and green hydrogen. By 2040 the target is to use 70% low-emissions hydrogen (i.e. blue and green)

Source: South Korean hydrogen roadmap; MOTIE; Ifri 2018; Green Car Congress; Press search

1.2 / South Korea focuses on scaling up domestic production of transport and electricity segments

NOT EXHAUSTIVE

x % share target of low-carbon hydrogen

	2018	2022	2040
Hydrogen[1] consumption, million tons per year	0.1	0.5	5 (70%)
Infrastructure roadmap	14 **refueling stations** 500 **tube trailers** 200km of **pipelines**	310 **refueling stations** Introduction of **high-pressure trailers for transport,** and large-scale **gas storage and transportation** Establish **pipelines near by-product H_2 production hubs** Build infrastructure for **overseas production** (from 2022)	1200 **refueling stations** **Liquefaction and liquid and solid storage and transportation** (2030) Construction of a nationwide **high-pressure pipeline network** (2030) Establish **nationwide supply** infrastructure (2030)
Targeted end-use sectors	**Transport:** 1.8k FC passenger vehicles production capacity, 2 buses	**Transport:** 79k FC passenger vehicles production capacity, 2k buses **Centralized power generation:** 1.5 GW **Decentralized power generation** (i.e. home fuel cells)**:** 50 MW	**Transport:** 5.9 Mn FC passenger vehicles production capacity, 60k buses, 120k taxi fleet, 120k trucks. Domestic and for export[2]. Ships, trains, and drones **Centralized power generation:** 15 GW **Decentralized power generation:** 2.1+ GW
Selected support schemes	**Subsidies for FCEVs purchase to incentivize domestic production of vehicles** at the beginning of the strategy (no year specified). Additionally, the government will also provide funding for the initial expansion of **refueling stations** and establish **more relaxed permit restrictions.** Support fuel cells deployment for power generation by my means of **weighted Renewable Energy Certificates.** Provision of financial incentives for **home fuel cells** such as the establishment of an **LNG-exclusive tariff** by 2022		

1. Includes grey, blue, and green hydrogen. Includes domestic production and imports; 2. In total, 3.3 Mn units for export and 2.9 Mn units for domestic consumption.

Source: Ministry of Economic Affairs and Climate Policy of the Netherlands; IEA; International Hydrogen Strategies (World Energy Council Germany); Hydrogen Economy Roadmap of Korea

1.2 / Key actions in for the creation of the hydrogen industrial ecosystem in South Korea – Enabling environment

NOT EXHAUSTIVE

Domain	Key measures	Description
Vision and targets	**Fostering of hydrogen industry**	Focus on creating a new industrial ecosystem and placing South Korea as a global leader in hydrogen utilization, by establishing clear targets, mainly in the **transport and power** generation sectors Support all stages of technological development throughout **the entire value chain**, from development of source technology in hydrogen-powered vehicles, core components of fuel cells, and storage and transportation, to demonstration, commercialization, and improvement of stability Develop guidebooks on **hydrogen safety** that address public concerns and include them in the **school curricula,** **Designate a 'Hydrogen Day'** and hold an exhibit with new technologies and programs
Regulation and licensing	**Promoting growth across the entire value chain**	**Start a detailed roadmap** for supporting the technological development throughout the **entire value chain**, in collaboration between the Ministry of Industry, Ministry of Science and Technology, Ministry of Land, and the Ministry of Water Resources. In addition, **launch a cross-ministerial preliminary feasibility study** for hydrogen (2021 to 2030) Establish legal **safety standards** to ensure the reliability and stability of the hydrogen economy throughout the entire value chain. Enactment of a "**Hydrogen Economy Act"** to provide an institutional base in H2 2019 Provide **relaxed licensing standards** for transportation companies introducing eco-friendly vehicles like hydrogen buses
Coordination and partnerships	**Enhancing international cooperation and industry alliances**	Lead international standardization and evaluate at least 15 cases **technical standards in hydrogen production, storage vessels, fueling systems, and fuel cells** by 2030 Establish a **cooperation system among Korea, China, and Japan,** and reinforce participation **in global partnerships** (e.g. Hydrogen Council) Enhance partnerships for the **development of overseas import base** from e.g. Middle East and Central and South America Promote a hydrogen economy in South Korea through **industry alliances** and international partnerships • **HyNet** (e.g. Hyundai, Air Liquide) promotes the establishment of hydrogen refueling infrastructure • **H2Korea** and **FCHEA** sign MoU in Feb. 2020 to increase collaboration in the hydrogen economy

Source: Ministry of Economic Affairs and Climate Policy of the Netherlands; IEA; International Hydrogen Strategies (World Energy Council Germany); Hydrogen Economy Roadmap of Korea

1.2 / Key actions in for the creation of the hydrogen industrial ecosystem in South Korea – Investments and incentives

NOT EXHAUSTIVE

Domain	Key measures	Description
Financing and incentives	**Securing the self-proliferation of the technology**	Offer **subsidies for FCEVs purchase** to incentivize domestic production of vehicles and subsidies for the **operation of refueling stations** as well as establish **more relaxed permit restrictions** Support fuel cells deployment for power generation by my means of **weighted Renewable Energy Certificates.** Provide financial incentives for **home fuel cells** such as the establishment of an **LNG-exclusive tariff** by 2022 Support FCEV business partners with long-term, **low-interest policy loans and investment expenses** Promote financial support to **establish mass production facilities**
Infrastructure	**Fostering technology maturity and expansion**	Establish a supply infrastructure network **nationwide by 2030** Convert from high-pressure gas storage to high-efficiency **liquefaction and liquid and solid storage** **Reduce cost of hydrogen distribution** by establishing high-efficiency, high-capacity hydrogen storage and transportation systems
R&D	**Performing large-scale demonstrations and supporting domestic production**	Create **a Hydrogen Industry Cluster** (2021) for R&D cooperation and for the development of a **large-scale testbeds** Help selected **companies to secure a global position** with R&D and finance support Build up large-scale demonstration projects for different end-uses. Selection of Ansan, Ulsan, and Wanju/Jeonju **as national testbed cities** by the South Korean Ministry of Land, Infrastructure, Transport, and Tourism. **Invest** to optimize the production of FCEV parts. Public funding for the investigation of **hydrogen trains** efficiency and stability, application of **fuel cells on ships**, and other transport-related activities **Start a pilot project for usage of hydrogen trucks** in the public sector, and a demonstration project for the conversion of general freight trucks
Skilled labor	**Training of safety standards and development specialists**	Create a **safety professional qualification** to manage safety across all processes of the hydrogen value chain Establish **a training program** covering the **design, operation, and safety of hydrogen refueling stations,** for station managers. **Train core personnel** on the design, production, and maintenance of **fuel cells,** Support SMEs with **research personnel and recruitment**

Source: Ministry of Economic Affairs and Climate Policy of the Netherlands; IEA; International Hydrogen Strategies (World Energy Council Germany); Hydrogen Economy Roadmap of Korea

1.2 / Dutch Hydrogen Strategy

NOT EXHAUSTIVE

General approach

The **h2 roadmap until 2050** was released in the Dutch **"National Climate Agreement"** in 2019, and further specific measures were introduced in the **"Government Strategy on Hydrogen"** in 2020

The strategy places hydrogen as a key enabler in the **long-term energy transition** and includes **specific targets in transport,** being other sectors' roadmap subject to further analysis and cost reductions

Key actions include government support through new and existing support schemes, international and public-private cooperation, and R&D

Additionally, the Dutch government launched the **Hydrogen Valley initiative** to **establish** a **hydrogen industry,** as response to **declining gas industry** in **Northern regions** Groningen, Drenthe and Friesland. Financed with EUR 20 Mn (~MUSD 23.4) by the EU, and EUR 70 Mn (~MUSD 82) by a consortium.

Currently there are **9+ GW of hydrogen projects announced**, ramping up towards 2040

Overview of key action areas

Regulation and licensing	Implement **safety standards and a policy framework** for hydrogen safety risks
Coordination and partnerships	Promote consultations and **bilateral cooperation with North Sea countries** to exploit the significant potential of offshore wind energy
R & D	Support H2 R&D projects through different programs such as **MOOI² tenders** (EUR 17 M) **and DEI+ program**, (EUR 15 M per project)
Skilled labor	Comprehensive framework to tackle labor issues in the **National Climate Agreement** including specific regional agendas for education development and labor market improvement

Key commitments and targets

0.5 GW of installed electrolysis capacity until 2025

3-4 GW of installed electrolysis capacity until 2030

100% climate-neutral economy by 2050

Source: Dutch Government Strategy on Hydrogen, Government publications; International Hydrogen Strategies (World Energy Council Germany)

1.2 / Dutch Hydrogen Strategy

	2025	2030	2050
Electrolysis capacity targets, GW	0.5	3-4	?
Infrastructure roadmap	50 **refueling stations. 10% investment costs reduction per year**	N/A	N/A
	Port of Rotterdam as **international hub for hydrogen import** and **distribution across Europe** (no timeline specified)		
	Evaluation of the necessity for **grid strengthening** and of an obligation for **hydrogen blending** into the **gas grid** (no year specified)		
Targeted end-use sectors	**Transport:** 15k FCEV, 3k HDV	**Transport:** 300k FCEV	**Aviation fuels:** 100% blending obligation
	Shipping: increase use of hydrogen in shipping industry (no year specified)	**Aviation fuels:** 14% blending obligation	**Power generation:** long-term relevance of hydrogen (no specific targets/plan provided)
	Industry: upgrading and refineries mentioned but not concrete plans stated		**Building sector:** higher relevance after 2030 due to uncertain technology costs
Selected support schemes	For a **transitional period,** to enable scaling up, **operational cost support** will be available **starting 2021** with an annual budget of **EUR 35 Mn**[1]		
	Starting in **2020,** the existing **SDE++** scheme[2] for the generation of sustainable energy will **include electrolysis**		
	Development of a system of **Guarantees of Origin** for **green H_2**		
	Allocation of EU funds through the **H2 Platform**		

1. The funds are made available through rearranging part of the existing DEI+ funds; 2. In total 2,000 full load hours are eligible for subsidy through this scheme with a maximum subsidy amounting to 300 EUR/ton abated CO_2

Source: Dutch Government Strategy on Hydrogen, Government publications; International Hydrogen Strategies (World Energy Council Germany)

1.2 / Dutch Hydrogen Strategy – Enabling environment

NOT EXHAUSTIVE

Domain	Key measures	Description
Vision and targets	**Introducing a vision for the future based on guidance and phasing**	Establish generation and sector targets in addition to those presented in **the Dutch National Climate Agreement.** The focus is on the following sectors: ports and industry clusters, transport, built environment, electricity sector, and agricultural sector Retain the current strategic position as an energy hub Foster cost reductions in order to **unlock scaling up of hydrogen production** and the subsequent set up of the supply chain, by means of new and existing support schemes
Regulation and licensing	**Developing policy tools to facilitate deployment in collaboration with the private sector**	Introduce a policy agenda based on **4 pillars**: legislation and regulation, cost reduction and scaling up green hydrogen, sustainability on final consumption, and a supporting and flanking policy Formulate a **National Hydrogen Programme** as part of the **National Climate Agreement,** based on the hydrogen phasing plan leading up to 2030 Review the formulation of a position on the **regulation of the value chain**. The main goals are to facilitate the market, to ensure security of supply, and to keep social costs the lowest possible Implement **safety standards** based on international and European guidelines, and implement a **policy framework for hydrogen safety risks** Launch a public-private **Hydrogen Safety Innovation Programme** to address safety issues
Coordination and partnerships	**Strenghtening European cooperation**	Communicate directly with the **European Commission** regarding EU hydrogen policies Start initiative, together with Austria, to develop common approaches to critical issues such as **standards, market incentives and market regulations,** in a Forum with **Benelux, Germany, France, Austria and Switzerland** Promote consultations and bilateral cooperation with **North Sea countries** to exploit the significant potential of **offshore wind energy** as key source for the production of green hydrogen beyond 2030. Focus on a strong European competitive position internationally through the IPCEI[1]

1. Important Projects of Common European Interest

Source: Dutch Government Strategy on Hydrogen; International Hydrogen Strategies (World Energy Council Germany);

1.2 / Dutch Hydrogen Strategy – Investments and incentives

NOT EXHAUSTIVE

Domain	Key measures	Description
Financing and incentives	**Offering financial support at the various phases of the development process**	Develop a reliable system of **Guarantees of Origin (GOs) and certification** Introduce a new and temporary **support scheme for operating costs related to the scaling up and cost reduction** process for green hydrogen Facilitate the scaling-up by making use of the existing **Climate Budget funds** as of 2021, with an annual budget of **EUR 35 Mn** Include electrolysis hydrogen production in the **SDE++ support scheme**[1]. Blue hydrogen production will be able to compete in the CCS category Explore the feasibility of a **blending obligation** of hydrogen in the distribution networks
Infrastructure	**Steering the deployment in collaboration with industrial players**	Review the **use of the existing gas grid,** and determine whether and under which conditions it could be used for hydrogen Coordinate the optimal placement of hydrogen infrastructure through the **Main Energy Infrastructure Programme** Focus on infrastructure as **key enabler for industry clusters** sustainability improvements
R&D	**Offering support schemes for research, scaling up and rolling out**	Support hydrogen projects through annual MOOI[2] tenders. In 2020, hydrogen projects are eligible to compete for a **EUR 17 Mn budget** (aprox. MUSD 20)**,** with a maximum **individual subsidy of EUR 4 Mn** (aprox. MUSD 4.7) Support R&D projects in an **industrial environment** through the DEI+ program, which covers **25-45% of eligible cost up to EUR 15 Mn per project** (aprox. 17.6 MUSD) Focus on **applied research** in collaboration with the business community through, among others, the **Netherlands Organization for Applied Scientific Research (TNO)** Research production and applications of green hydrogen through the **Energy Top Sector** as part of the various multi-year mission-driven innovation programs
Skilled labor	**N/A**	N/A

1. Support scheme for cost-effective carbon emissions reduction

Source: Dutch Government Strategy on Hydrogen; International Hydrogen Strategies (World Energy Council Germany);

1.2 / German government agrees on National Hydrogen Strategy

Description

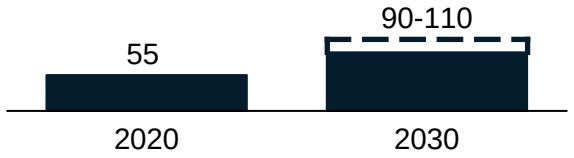
German Government agrees on **National Hydrogen Strategy** that aims at providing a **framework for future production** and **application of hydrogen** as well as related **innovations and investments**

Long-term focus is on **green** production, however, the role of **other low-carbon pathways is acknowledged** as import tool for **short- to mid-term** market acceleration

The strategy aims at laying the **foundation** for a **functioning hydrogen market** until **2023**

The government and a **National Hydrogen Council** will **evaluate and review the strategy** every **3 years**

Demand for hydrogen in Germany, TWh



Overview of key measures

The strategy includes **38 measures** that aim at establishing a value chain for the **production, distribution, and large-scale use of hydrogen**

Generation	Support **green hydrogen production** via new **support schemes** (contracts-for-difference, auctions) and **investment grants**
Infrastructure	Plan **supply infrastructure** via **dedicated pipelines** and **existing gas networks**
Hydrogen usage	Support **usage** of hydrogen in **various sectors** (e.g., industry, transport, heating) via dedicated **support schemes** and **regulations,** and industry-specific dialogues Define **standards for hydrogen products** to establish **international market**
Partnerships	Intensify **international partnership projects** to enable **imports of green hydrogen** as **domestic production** (14 TWh of in 2030) is expected to **be insufficient for deep decarbonization**

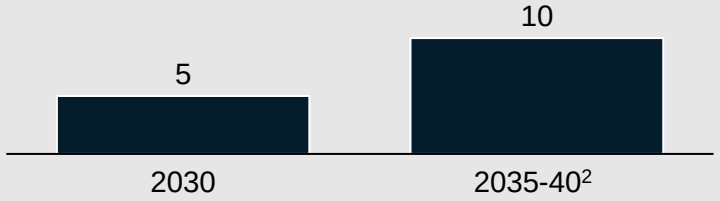
Key commitments and targets

7 bn EUR investments for **domestic hydrogen market ramp-up**

2 bn EUR investments for hydrogen projects within **international partnerships**

2% quota for **green hydrogen-based aviation fuels** until 2030 to be investigated

Installed electrolysis capacity targets [1], GW



1. The strategy mentions a target of 2 GW for electrolyzer but does not specify a concrete time frame or if this is additional to the 5-10 GW targets
2. Additional 5 GW to be added until 2035 if possible but no later than 2040

Source: Nationale Wasserstroffstrategie (National Hydrogen Strategy), BMWi

1.2 / Germany's national hydrogen strategy gets continuously revised in 3-year cycle

x Renewable hydrogen production, TWh/yr

	Phase 1: 2020-2023	Phase 2: 2023-2030	Beyond 2030
Objective	**Market launch**– Decarbonize existing hydrogen production and facilitating take up of hydrogen consumption in new end-uses	**Scale up** – Expand production and end-use to additional sectors to develop hydrogen as intrinsic part of an integrated energy system	Use hydrogen to establish CO2-neutral economy by 2050
Electrolysis capacity targets, GW		5 (14) — 2030	10 (28) — 2035-2040
Infrastructure roadmap	Formulate report on **recommendations** for hydrogen supply through **dedicated pipelines** as well as **hydrogen-readiness of existing gas infrastructure,** and **buildout of hydrogen gas stations** **Redesign planning and financing** of infrastructure (electricity, heat and gas) to allow for **cross-coordination and cost-efficient deployment** of energy transition capable energy infrastructure Create hydrogen refueling station infrastructure		
Targeted end-use sectors	**Transport:** 3.6 bn EUR in subsidies for vehicles and 1.1 bn for fuels until 2023[1] **Heating:** Support hydrogen fuel cell heating **Industry:** Subsidize hydrogen use in chemicals and steel. 55 TWh/yr consumption (2020)	**Transport:** 2% green hydrogen in aviation fuel **Heating:** Further promote hydrogen use **Industry:** >65TWh/yr consumption	
Selected support schemes	**Subsidies** from energy- and climate fonds **Exclude green hydrogen** from **EEG-levy** Grant **investment support** for **electrolyzers** to support industry **shift from grey to green hydrogen** Establish pilot project where generators receive difference between **CO2 abatement costs for producing green hydrogen** (fixed CO2 price) and the **CO2 price in the EU emission trading system** to guarantee generators stable returns[4]		

1. Including electric vehicles and biofuels

Source: Nationale Wasserstroffstrategie (National Hydrogen Strategy), BMWi

1.2 / Key actions in for the creation of the hydrogen industrial ecosystem in Germany – Enabling environment

NOT EXHAUSTIVE

Domain	Key measures	Description
Vision and targets	**Ramping up the green hydrogen market and using it to enable Germany's energy transition**	Create a **global impact on GHG reduction** by developing a **market for green hydrogen** and increase **global cooperation** Enable **hydrogen** as a **competitive source of energy** by driving technological development and advancing cost degression, establish **green hydrogen** in **industrial processes** and as a **fuel source** for seafaring and aviation Establish a **domestic market** that **incentivizes hydrogen** use and develop hydrogen infrastructure to take advantage of **synergies with the energy transition** in Germany Improve the **hydrogen framework and R&D** continuously
Regulation and licensing	**Promoting the growth of the entire value chain**	**Implement** the continuously revised **hydrogen strategy** **Regulatory unbundling** of **electrolyzers** and power and gas **grid operators** Develop and improve **framework conditions** for pairing **offshore wind** and electrolyzers including the designation of **dedicated land/areas and the possibility of additional tenders** Start a **European dialogue** between regulators and the industry on **decarbonization strategies**, including hydrogen as an option
Coordination and partnerships	**Preceding International Standards and promote industry alliances**	Develop **joint** European **regulations** and **industry standards** to establish **cross-border hydrogen market** and support **investments** in hydrogen technologies via **„Important Project of Common European Interest (IPCEI)“** Integrate hydrogen in **international energy partnerships** to jointly develop **hydrogen technologies** and thereby guarantee that Germany’s **future demand** for hydrogen can be met **via imports** Advance **standardization** of hydrogen systems in **mobility**

Source: Nationale Wasserstroffstrategie (National Hydrogen Strategy), BMWi

1.2 / Key actions in for the creation of the hydrogen industrial ecosystem in Germany – Investments and incentives

NOT EXHAUSTIVE

Domain	Key measures	Description
Financing and incentives	**Establishing green hydrogen as a competitive source of energy**	**Subsidies** from energy- and climate fonds **Exclude green hydrogen** from **EEG-levy** Grant **investment support** for **electrolyzers** to support industry **shift from grey to green hydrogen** Establish pilot project where generators receive difference between **CO2 abatement costs for producing green hydrogen** (fixed CO2 price) and the **CO2 price in the EU emission trading system** to guarantee generators stable returns[4]
Infrastructure	**Developing a reliable and secure infrastructure**	Formulate report on **recommendations** for hydrogen supply through **dedicated pipelines** as well as **hydrogen-readiness of existing gas infrastructure,** and **buildout of hydrogen gas stations** **Redesign planning and financing** of infrastructure (electricity, heat and gas) to allow for **cross-coordination and cost-efficient deployment** of energy transition capable energy infrastructure Create hydrogen refueling station infrastructure
R&D	**Promoting research and innovation in hydrogen technologies**	Strategically **bundle multiple research initiatives** towards hydrogen and investigate measures to facilitate **market entries** for **new hydrogen technologies** Provide **EUR 25 million funding for hydrogen research** in **aviation and shipping respectively** over the 2020-2024 period
Skilled labor	**Training of safety and development specialists**	Find **new ways of collaboration** between **research** and **education** to **maintain** and **expand qualified workforce** **Support** and promote **training and development** of workforce **in hydrogen**, especially in production, processing and maintenance

Source: Nationale Wasserstroffstrategie (National Hydrogen Strategy), BMWi

Chapter 2: Business case for domestic Hydrogen production and end use application

Chapter content description

- **Business cases for domestic production of green hydrogen** through green renewables, including costs of generation and hydrogen production
- **Business case for prioritized end use applications**[1] **including:** heavy transport, urban buses, mining trucks (CAEX), natural gas blending, industry heating, green ammonia production and replacement of grey hydrogen
- **Domestic demand projections for green hydrogen in Chile** by end use application
- **Identify enabling factors** with the greatest influence over the business case

Activities included

- **Activity 2.1**
- **Activity 2.2**
- **Activity 2.3**
- **Activity 2.4**

1. End use applications prioritized based on market potential and impact on strategic roadmap and do not represent an exhaustive list. Some end use applications such as ´energy storage´ were included in the original scope of work but deprioritized in favor of replacement of grey hydrogen with green hydrogen, an end use application with the potential to reach economic break even in the short term.

2.1/ To model the potential of Green Hydrogen in Chile, we need to optimize 3 factors of production costs

Generation cost

Generation location

Capacity Factor sequencing

Generation technology (wind, PV, Bifacial and CSP)

Electrolyzer utilization

RES blending

Storage technology

Transmission / distribution

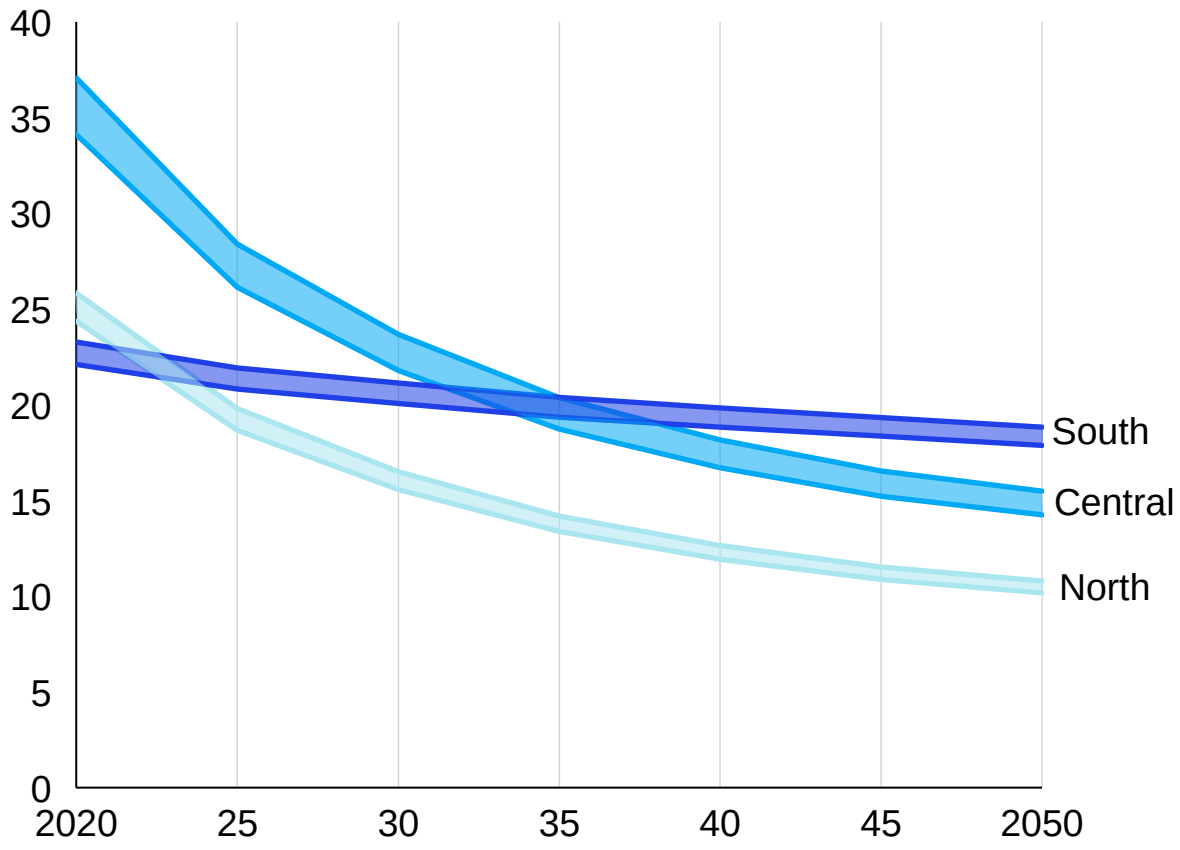
H2 production location

Transmission costs

Transportation costs

2.1/ In the three macro-zones considered, LCOE costs in the northern and southern zones can reach as low as ~19 USD / MWh and ~21 USD / MWh respectively by 2025

Levelized cost of energy[1] for competitive zones, USD/MWh



Capacity factors by region[2], %

	Competitive zones		Uncompetitive zones
Northern Production Zone	Atacama PV	34%	Coquimbo Wind
	Calama PV	33%	Taltal Wind
	Taltal PV	35%	Calama Wind
Central Production Zone	Central PV	23-25%	Chiloe Wind
			Biobio Wind
			Coquimbo Wind
Southern Production Zone	Magallanes wind	56-59%	Bahía Inútil
			Offshore wind

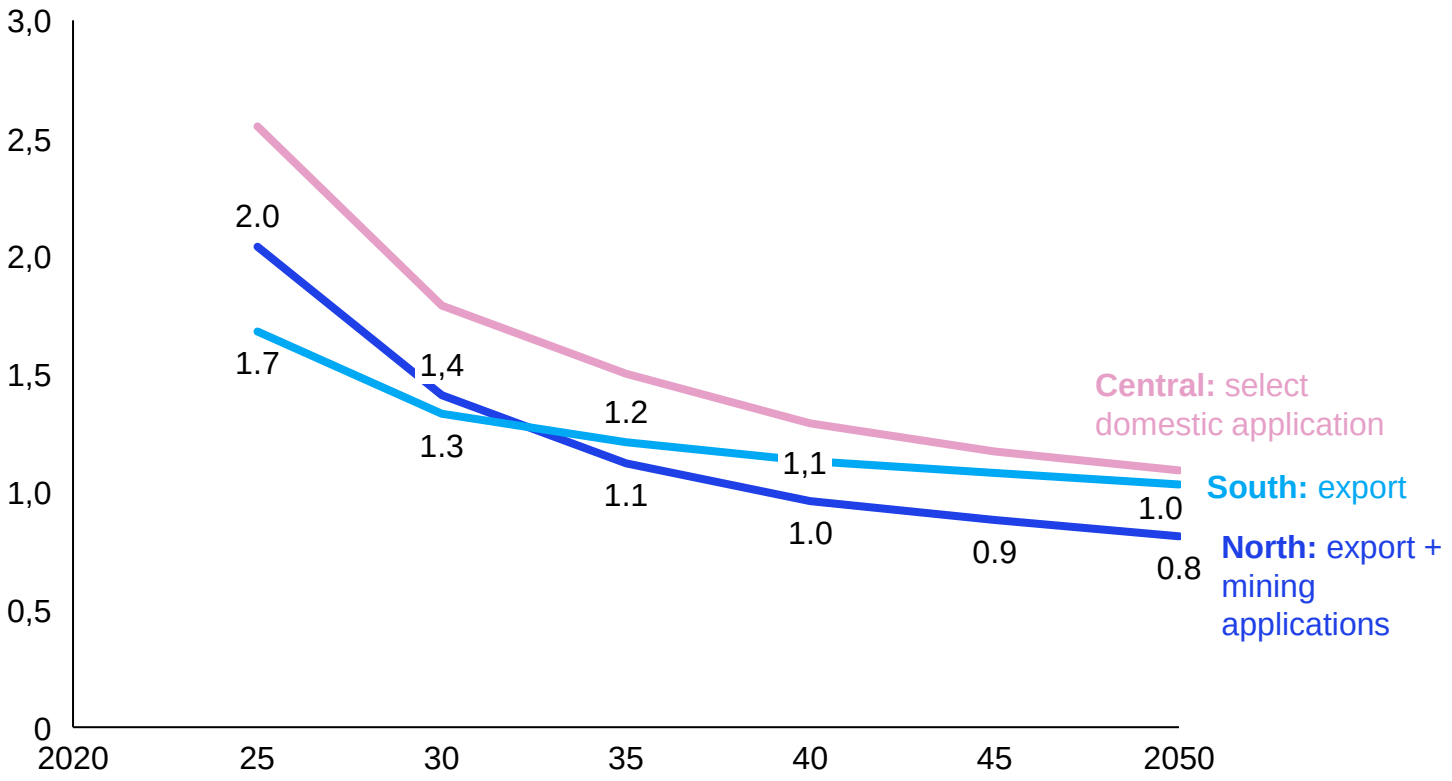
Competitiveness still being assessed

1. Based on 6% WACC, does not include transmission costs
2. Capacity Factors for Tier 1 sites

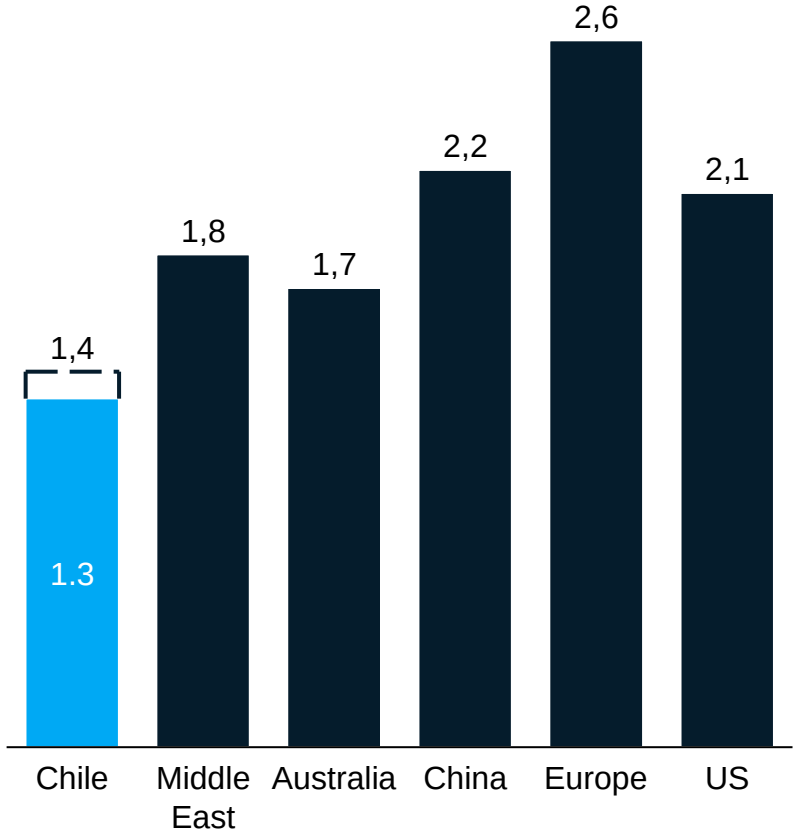
Source: McKinsey Hydrogen Cost Model

2.1/ Chile´s unparalleled renewable resources in the Atacama and Patagonia makes it the lowest cost place to produce Green Hydrogen in the world

Production cost curve for hydrogen by region,
Generation and Electrolyzer costs of LCOH, USD / kg H2



Comparison of production costs 2030
Generation and Electrolyzer costs of LCOH
USD / kg H2

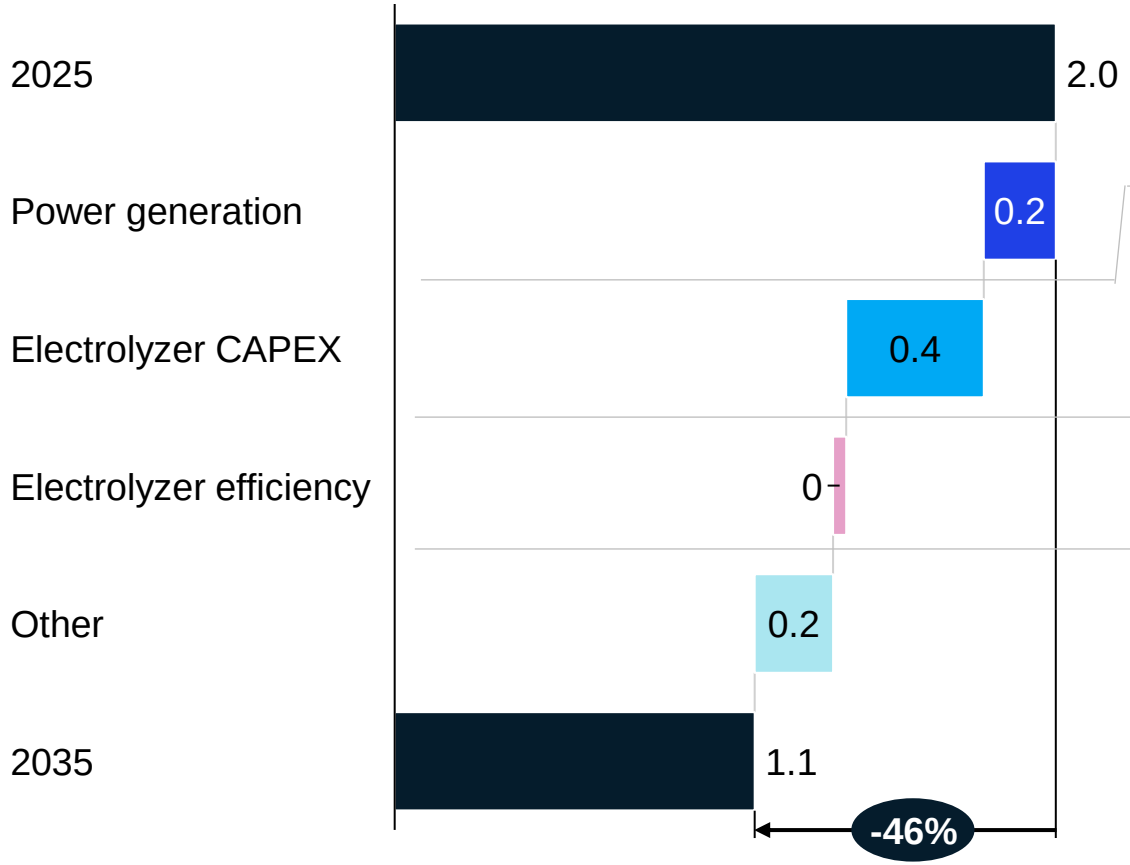


Source: McKinsey model, Minister of Energy capacity factor data and PV / Solar profiles

2.1/ Green hydrogen production cost is projected to drop by 50% within the next 10 years driven by a decrease in electrolyzer costs

Cost reduction levers for hydrogen electrolysis connected to dedicated solar PV in Taltal, Chile

USD/kg H_2



Power generation costs decrease by 1/3, driven by a steep decline in solar PV CAPEX from ~850 $/kW to ~400 $/kW. Solar PV LCOE in Taltal, the most promising area in the North, decreases from 19 to 13 USD/MWh

Electrolyzer CAPEX costs decrease sharply by 55% in 10 years. This is due to a combination of factors, including: larger global production scale, automation in manufacturing, larger individual system size classes (e.g., going from 2-20 MW today to >100 MW per unit), decreasing BOS[1] costs,

Electrolyzer efficiency increases from 68 to 71% due to technical advancements, such as better catalysts

Other costs go down, especially electrolyzer O&M costs, following a reduction in the cost of parts and learning to operate systems

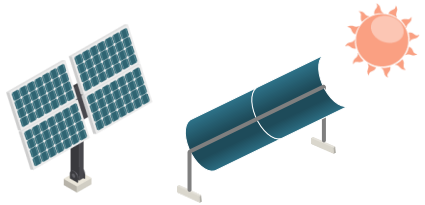
1. Balance of system

Source: McKinsey Hydrogen Cost Model

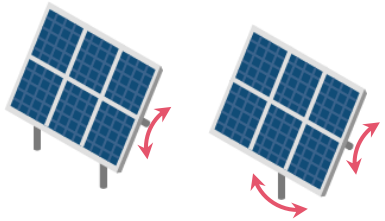
2.1/ Different options for solar technology range from fundamental changes to incremental improvements to the system

Details to follow

Fundamental shifts on how the system works



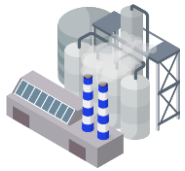
PV vs CSP[1]



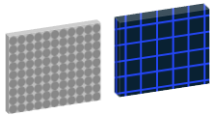
PV 1-axis vs 2-axes tracking

Battery storage

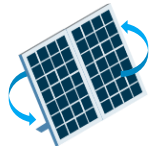
Incremental improvements to PV technology



Manufacturing scale



Single vs multi-crystalline Silicon wafers



Bifacial modules

Perovskite solar cells

Professionalization of the industry

Compared to PV technology, CSP represents a fundamental shift on how the system works, leading to significant differences in LCOE. By contrast, bifacial modules provide incremental improvements to PV technology, with a marginal LCOE change that is highly dependent on specific local characteristics, such as albedo (ground material) and labor cost (more complex engineering work is required for project planning and installation)

1. Concentrated Solar Power

2.2-2.3/ Economic breakeven will start with domestic applications; exports of green ammonia will ramp up in the second half of this decade, whereas exporting hydrogen and synfuels will happen after 2030

Not exhaustive **Preliminary**

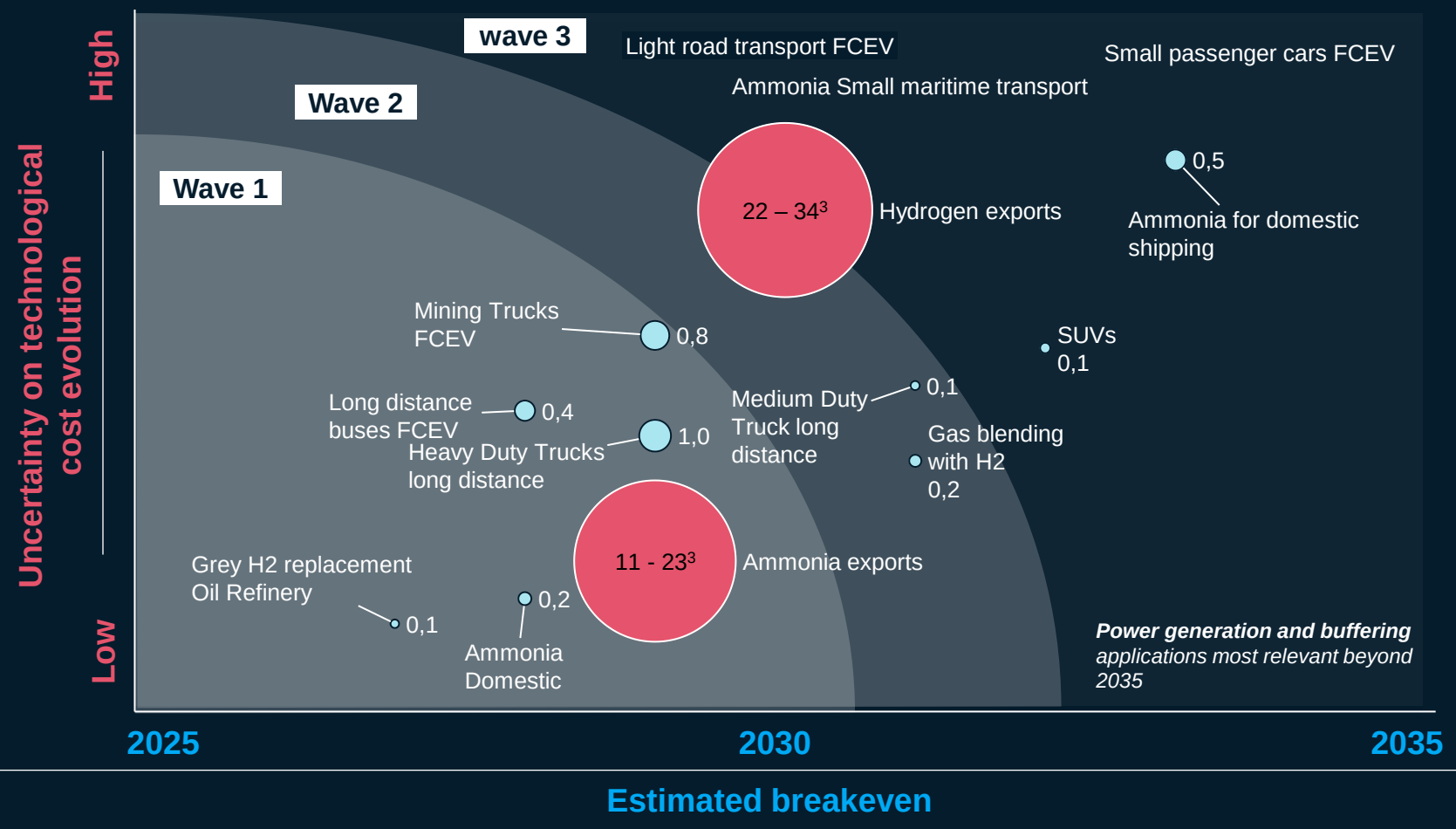
Prioritization matrix for main hydrogen applications[2]

Uncertainty vs Break-even – size of the bubble represents total potential market for 2030

1 GW Electrolyzer Capacity = ~0,1 Mton

5,00 – Mton hydrogen equivalent

Domestic International[1]



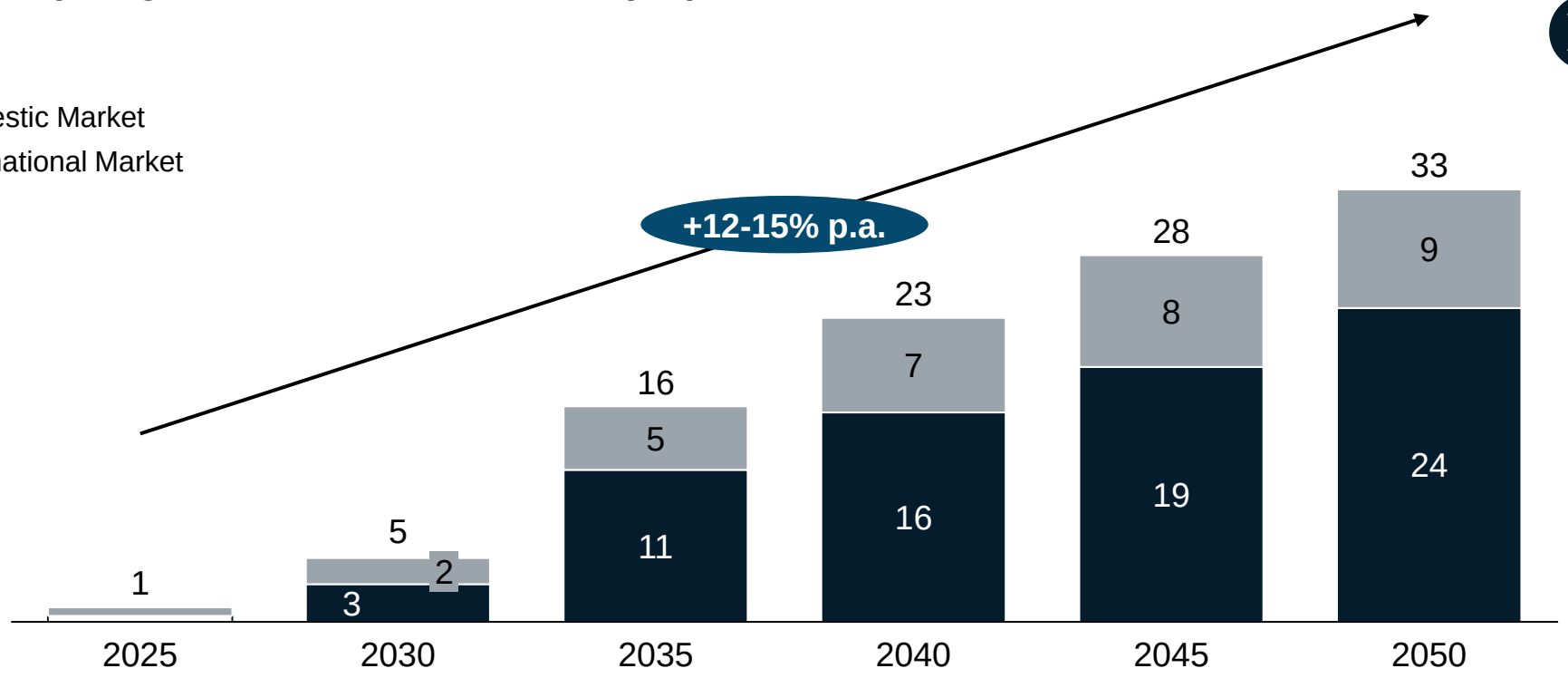
- Commercialization of hydrogen applications will take place in three waves:
  - Wave I: Domestic ramp up and export preparation
  - Wave II: Capitalization on export markets
  - Wave III: Leverage scale to expand
- Additional applications (not pictured) can become relevant once the synergies from large scale exports are captured
- Key longer term applications (synfuel and methanol) will bring additional market opportunities in 2035+
- Industrial head applications considered, but they are small

1. Considers domestic demand of ammonia and ammonia used for Fertilizers | 2. Does not consider CO2 Tax | 3: Considers main addressable markets (Low range does not consider China)
Source: Team Analysis; Hydrogen Council report; INE Chile; OCDE

2.2-2.3 / Market size of the hydrogen industry in Chile

Chilean Hydrogen and derivative market projections: 2025-2050

B USD



	2025	2030	2035	2040	2045	2050
Required GW electrolyzer capacity	3-5	25	90	130	155	190
Required GW RES capacity	5-8	40	145	200	250	300

However, capturing this opportunity requires the **appropriate ramp up of applications:**

Wave I: Domestic ramp up and export preparation

Wave II: Capitalize on export markets

Wave III: Leverage scale to expand

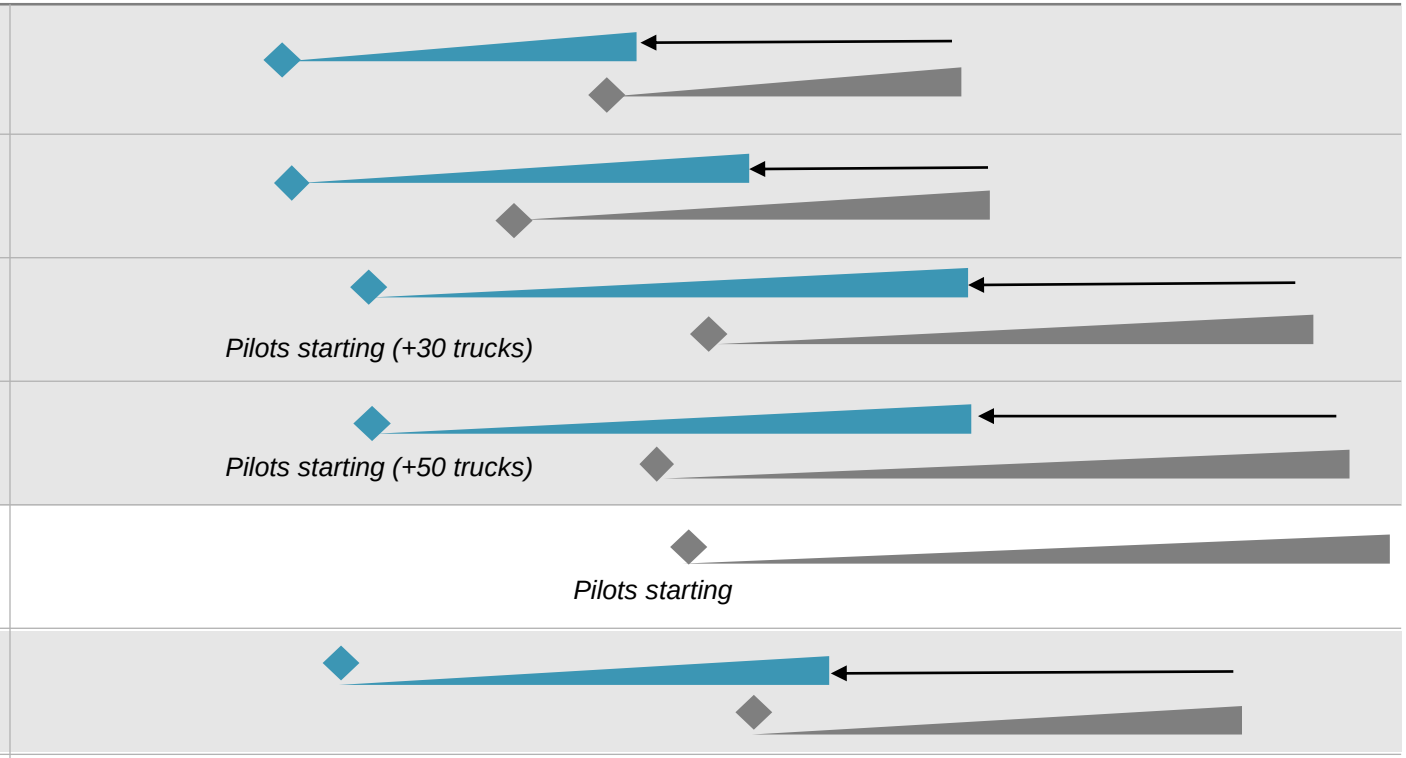
Detailed next

2.2-2.3 / Domestic application anticipation would be focused primarily on grey H2 replacements in oil refinery, grey NH3 replacement in mining / fertilizers and glass blending

Preliminary

Estimated break-even with National H2 Strategy | *Potential break-even without National H2 strategy* | *Potential ramp-up to achieve 100%*

	Application	Total Potential Market size[1] - USD B 2050	Electrolyzer capacity required (GW)
1	Current H2 replacement Oil refinery	0.2	1,8
2	Current NH3 replacement Mining + fertilizers	0.5	2,2
3	Mining haul trucks	1.6	9,3
4	Heavy duty trucks	2,0	11
5	Coaches / buses	0.8	4,5
6	Gas Blending (up to 20% total residential use)	0.3	1,8
		~6 USD B	



GW Electrolyzer: (w/ National H2 Strategy)	2025	2030	2035	2040
	3	10	25[2]	40[2]

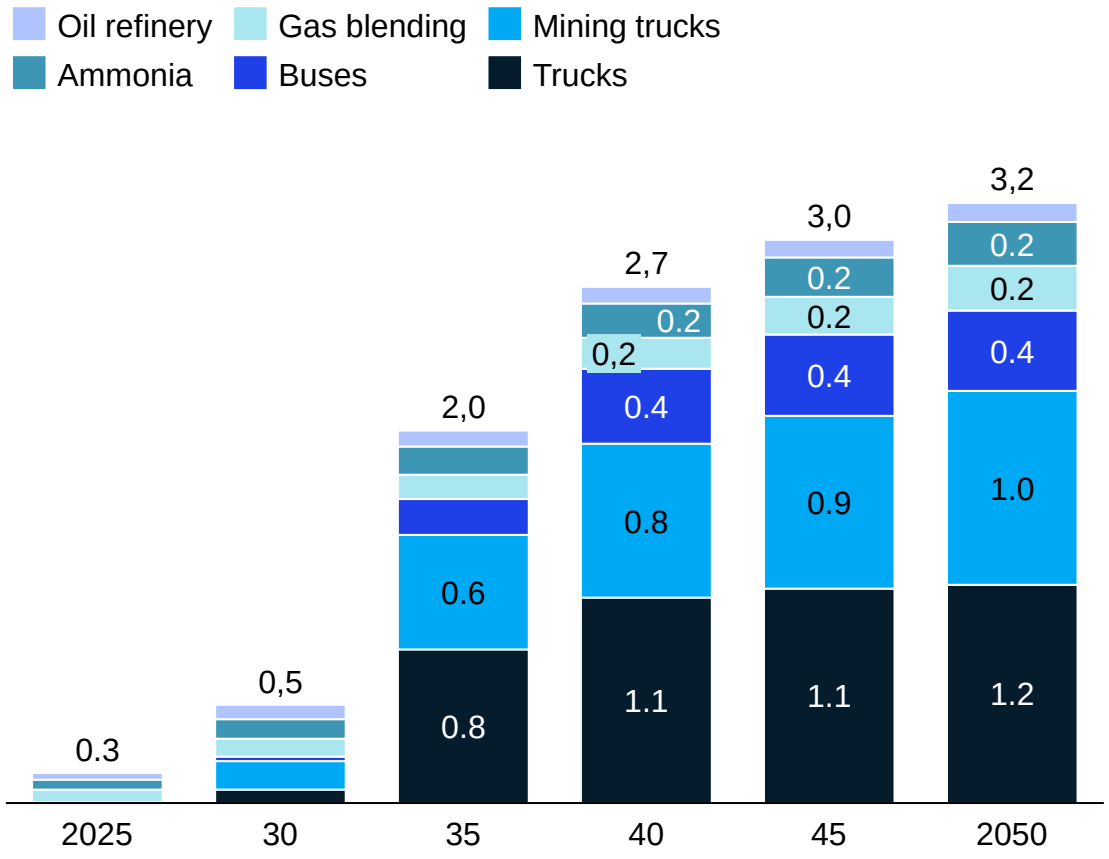
1. Consider the transition to hydrogen of 100% of the energy demand of each application
2. Including other domestic applications including: airlines, shipping and large passenger cars

Source: Team analysis; INE, ENAP; DPO

2.2-2.3 / Domestic could account for +3M ton of total hydrogen demand by 2050

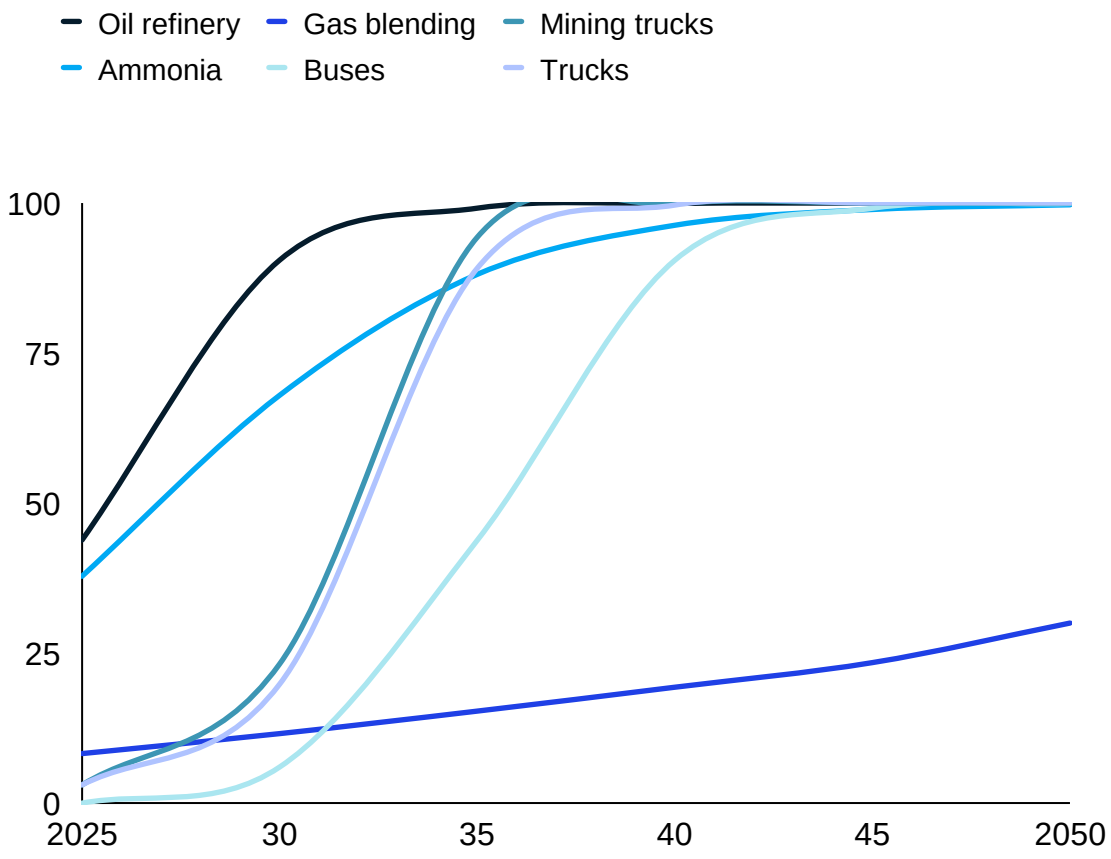
Hydrogen uptake for different domestic applications

Mtons of hydrogen equivalent



Hydrogen uptake for different domestic applications

% of total consumption



2.4/ Factors which produce the largest differences between the results of the business cases

1 Financing and incentives

Sources of funding to facilitate the launch of feasibility studies, pilots and scaled projects

Incentives structured to support supply and demand development for scaled projects

2 Regulation and licensing / permits

Long term regulatory structure to diminish uncertainty and minimizes barriers to entry / doing business

Acceleration of licensing and permitting processes

3 Coordination and partnerships

Coordination mechanism to create formalized partners with domestic and international actors to promote industry growth and domestic and international supply chain development

4 Infrastructure

Necessary infrastructure to support project development (transmission, ports, refueling stations etc…)

5 Research & Development

Research and development of technologies for the Chilean context (ex. wind turbines for Patagonian wind profiles and earthquake resistant ammonia production)

6 Skilled Labor

Ability to attract local talent for construction and production

Source: Press research; Hydrogen roadmap for EU, USA; Netherlands, France, Australia, Germany, South Korea; Team analysis

Chapter 3: Business case for Hydrogen exportation

Chapter content description

- **Demand projections for key export markets of green hydrogen** by end use application and geography (focus on green hydrogen and green ammonia)
- **Competitiveness assessment of Chile in key export markets, including**: European Union, Japan, South Korea, China and the United States
- **Key factors which influence the competitiveness of Chile in these export markets**

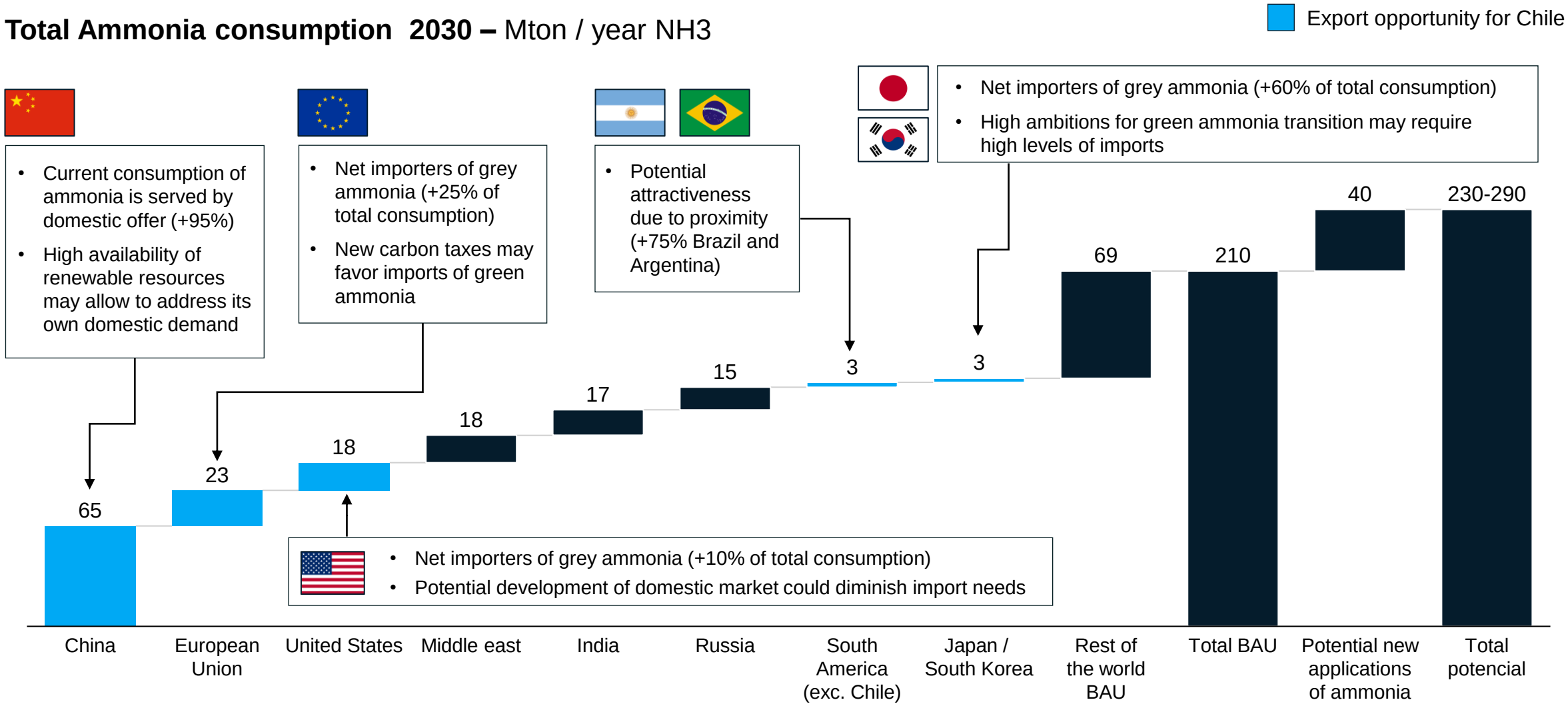
Activities included

- **Activity 3.1**
- **Activity 3.2**

1. End use applications prioritized based on market potential and impact on strategic roadmap and do not represent an exhaustive list. Some end use applications such as ´energy storage´ were included in the original scope of work but deprioritized in favor of replacement of grey hydrogen with green hydrogen, an end use application with the potential to reach economic break even in the short term.

3.1 / **Ammonia exports:** key potential green ammonia markets for Chile include China, Japan/South Korea, EU, USA and LATAM

Total Ammonia consumption 2030 – Mton / year NH3



3.1 / **Hydrogen export:** key hydrogen markets for Chile include South Korea, European Union and the United States

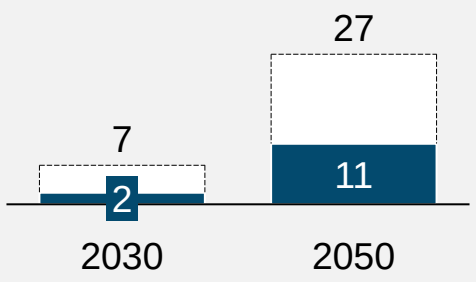
Preliminary

Potential demand for green hydrogen per country

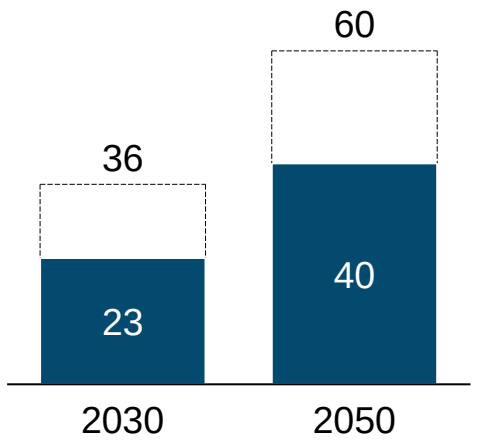
Mton H2 / year; Low and high ranges found in strategic roadmaps

Export opportunity for Chile | Ambitious scenario | Base scenario

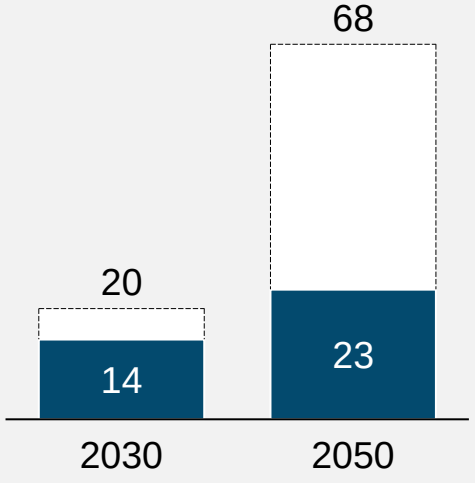
- Japan's Energy foresees the procurement of ~300 Ktons in 2030 and 5-10 Mtons in +2050
- S. Korea plans to start importing hydrogen in 2030



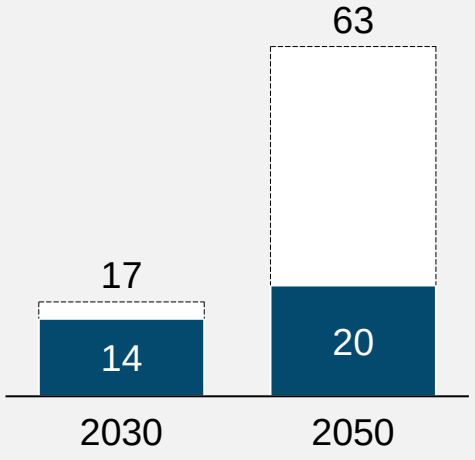
- Domestic production capacity may be sufficient to attend the demand for green hydrogen, however, accelerated expansion may require imports



- The limited production capacity in the region may boost imports specially by countries as Germany, Netherlands, Belgium, among others



- Even though the country counts on its own natural resources, Chilean hydrogen presents an attractive cost structure able to compete with the local market

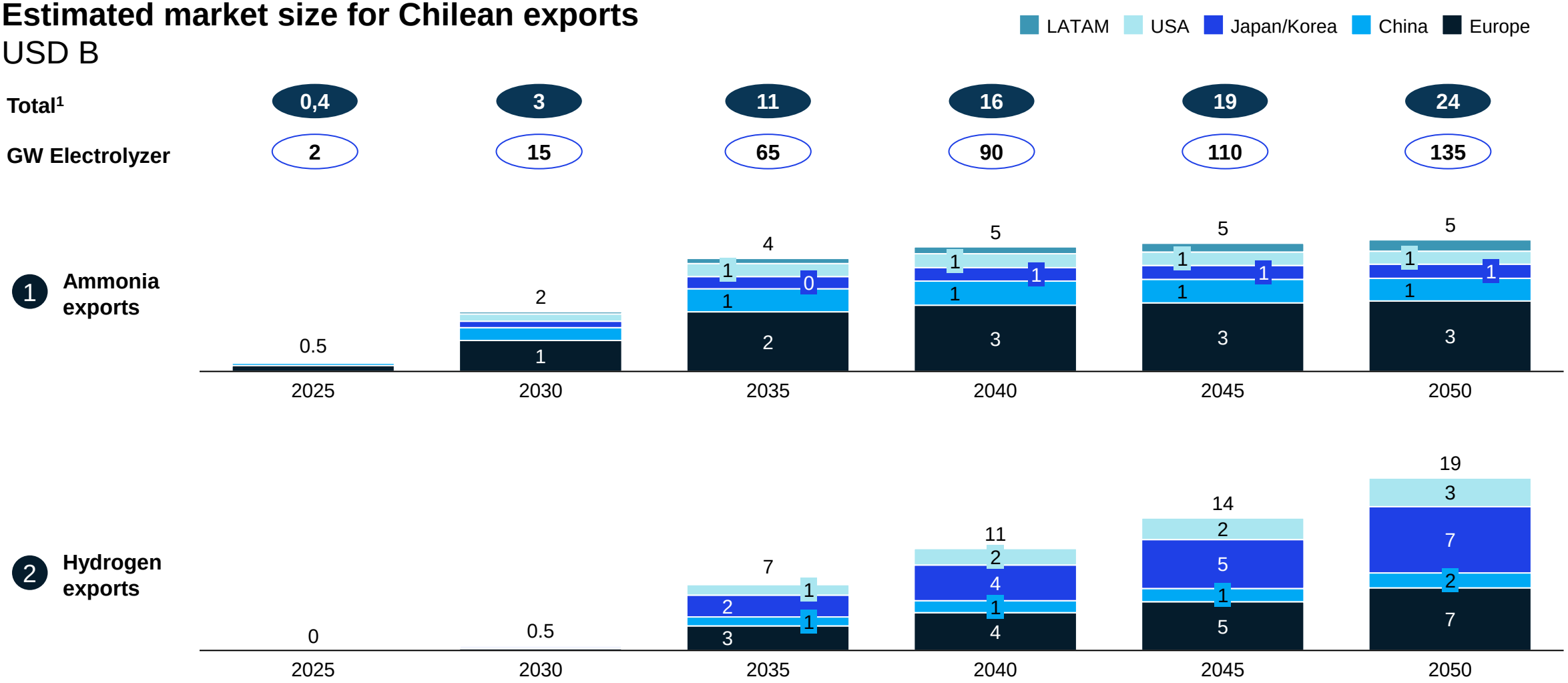


Source: China renewable energy outlook 2018; Hydrogen roadmap South Korea 2018; Japan Strategy Energy Plan 2018; Roadmap to a US hydrogen Economy; Perspectives on hydrogen for the APEC region; Hydrogen Council; EU strategic roadmap; The future of Hydrogen IAE;

3.1 / Export markets have the potential to ramp up starting 2025, provided that long term contracts are brokered with international players

Estimated market size for Chilean exports

USD B



1 Total of Ammonia + Hydrogen exports

Source: Team analysis; National hydrogen roadmaps of the respective countries -Total of Ammonia + Hydrogen exports

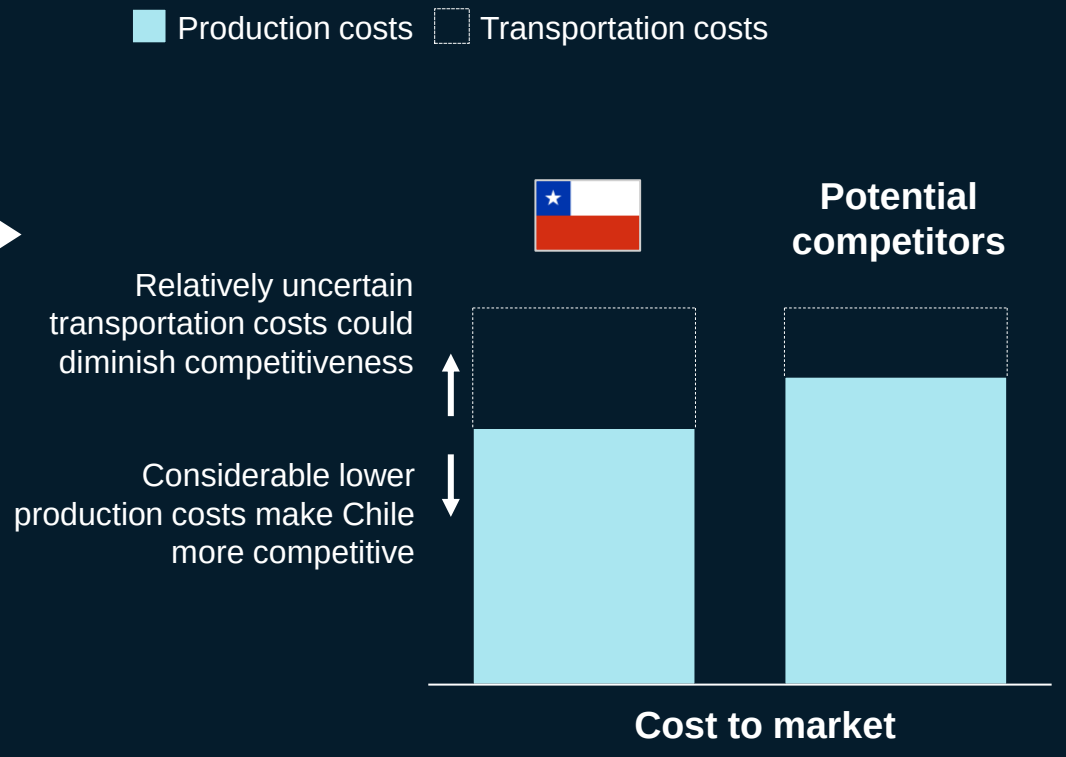
3.2 / Factor that will influence Chilean export competitiveness

1. **Hydrogen roadmaps and decarbonization targets of main net importer markets of green hydrogen and derivatives**

2. **Chile's production and transportation costs vs potential competitors to serve those markets**

3. **Acceleration of export production capacity development to capture potential market share**

Chile's strategic position will depend on the offset between lower production costs and higher transpiration costs



Chapter 4: hydrogen industry development targets and roadmap

Chapter content description

- **Hydrogen industry targets for National Green Hydrogen Strategy**
- **Roadmap for domestic application development in Chile**

Activities included

- **Activity 4.1**

1. End use applications prioritized based on market potential and impact on strategic roadmap and do not represent an exhaustive list. Some end use applications such as ´energy storage´ were included in the original scope of work but deprioritized in favor of replacement of grey hydrogen with green hydrogen, an end use application with the potential to reach economic break even in the short term.

4.1 / We aspire that Chile be among the top 5 producers of green hydrogen worldwide and the top 3 exporters; our target is 5 GW of hydrogen capacity under development by 2025, 25 GW by 2030

Hydrogen Roadmap

Aspiration:

Top 5

Global producers

Top 3

Global exporters

Targets: GW of electrolyzer capacity in development

5 GW

by 2025

25 GW

by 2030

Hydrogen valleys:

4.1 / Capturing this opportunity depends on the short term ramp up of domestic applications

Proposed sequencing and timing of application ramp up

Sequence of steps

Detailed next

Wave I: 2025

Domestic ramp up and export preparation

3-5[1] GWs

Kick start hydrogen industry with large domestic applications with earliest economic breakeven:

- Grey H2 replacement in Oil Refinery application
- Grey ammonia replacement
- Gas blending
- Pilots for mining Haul Trucks and Heavy Duty Trucks

Build knowledge, scale, and infrastructure to move into export opportunities (primarily Green Ammonia)

Wave II: 2030

Capitalize on export markets

25 GWs

Leverage domestic base to aggressively scale into a relevant player in the most attractive export markets:

First, Green Ammonia export market (principally: Latam, Europe, US and China)

Second, Green Hydrogen export market (principally: Europe, Japan / South Korea, maybe US and China)

Wave III: 2035+

Leverage scale to expand

90+ GWs

Synergies and economies of scale enable expansion into additional domestic applications

Scale up export to other markets to become a significant supplier of world energy consumption

Capture opportunities in future technologies and applications of Green Hydrogen and derivatives as they become economical:

- Synfuel applications (including Jetfuel)
- Ammonia for shipping

1. 3 GW for domestic applications and 2 GW additional based on accelerated export opportunities

Source: McKinsey analysis

4.1 / Optimal sequence and ramp up timeline of applications, corresponding infrastructure and skilled jobs required

Key milestone for the Chilean hydrogen strategy

▼ Pilot and infrastructure launch

◢ Application and infrastructure ramp up

		Preparation: Today - 2025	Capitalize exports: 2025 - 2030	Scale up and expansion : 2030 - onwards		
		Today	2025	2030	2035	2040
Production	Electrolyzer capacity (GW)		3 - 5	25	90	130
Application uptake – **Domestic**	H2 replac. (oil refinery)	Substitution of domestic uses ▼		Substitution of H2		
	NH3 replacement	Substitutions of exports ▼		Substitution of grey NH3		
	Mining Haul trucks	Initial bigger scale pilots ▼			+80% fleet renovation	
	Heavy duty trucks	Initial bigger scale pilots ▼			+80% fleet renovation	
	Coaches / Buses		Initial bigger scale pilots ▼			+80% fleet renovation
	Gas Blending	▼			+15% blending	
Application uptake – **Exports**	NH3 exports	◆ Bilateral partnership agreements	Initial exports ▼		+80% potential captured	
	H2 exports	◆ Bilateral partnership agreements		Initial exports ▼		+80% potential captured
Impact	Killed jobs required – Construction and industry operation		30 k	145 k	+400 k	+400 k

1. Includes investment in energy generation, electrolyzer, transmission and compression, transportation, liquefaction and refueling facilities

Source: DPO; TCO Model; team Analysis

Chapter 5: relevant stakeholders for the hydrogen industry

Chapter content description

- **Map of relevant stakeholders (private and public, domestic and international) who could have an important role in the development of a Chilean hydrogen ecosystem**
- **Identify enabling factors which could facilitate the engagement of these stakeholder groups in Chile´s developing hydrogen ecosystem**

Activities included

- **Activity 5.1**
- **Activity 5.2**
- **Activity 5.3**
- **Activity 5.4**

1. End use applications prioritized based on market potential and impact on strategic roadmap and do not represent an exhaustive list. Some end use applications such as ´energy storage´ were included in the original scope of work but deprioritized in favor of replacement of grey hydrogen with green hydrogen, an end use application with the potential to reach economic break even in the short term.

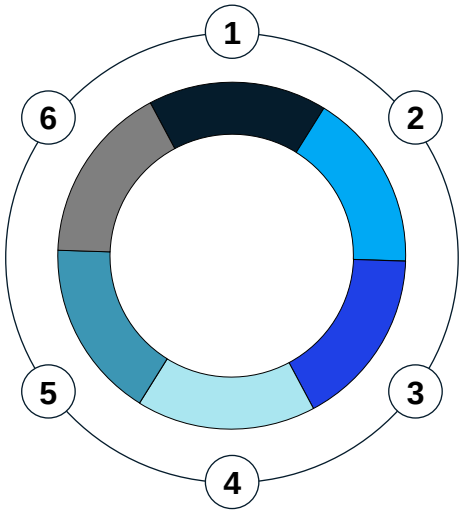
5.1 & 5.2/ The development of the Chilean hydrogen ecosystem will depend on 6 key groups of stakeholders

Off-taker group

Stakeholders that will diminish the risks of the projects guaranteeing volume demand; could be also co-investors

Ecosystem builders

Responsible for creating the enabling conditions for the hydrogen market



Financing investors

Funds or banks willing to invest in the energy market in Latin America

Hydrogen production

Responsible for technology and infrastructure development for hydrogen production, transportation and storage

Demand technology developers

Technology developers for different hydrogen applications – demand side (e.g. transportation)

Electricity generation (provider or developer)

Current or new energy generation players that will dedicate resources to hydrogen projects

Key considerations

- The coordinated actions of different stakeholder groups will allow to accelerate the growth of the hydrogen market by:
  - Facilitating the creation of the enabling environment for the market to operate
  - Rapidly acquiring and incorporating the hydrogen technology development to boost demand
  - Coordinating financing, energy generation and production promoting the development of new projects

5.1 & 5.2/ Stakeholder groups (1/3)

Not exhaustive

XX Possible champions

Group	Role(s)	Organizations
Potential off-takers 1	Off-takers of green hydrogen production for oil refinery	**ENAP**; Linde
	Off-takers of green hydrogen for mining trucks	**Codelco, BHP, Anglo-American**, Teck, **Antofagasta Minerals**, SQM, Barrick gold
	Off-takers of green hydrogen for inter-urban buses	**Turbus,** Pullman Bus
	Off-takers of green ammonia for explosives	**ENAEX**
	Off-takers of green hydrogen for gas blending	Metrogas, GasValpo, GasSur
	Off-takers of green hydrogen and green ammonia exports	**Yara, Mosaic;** Linde; ThyssenKrupp, APM Maersk, Air Liquid (Airgas), Casale
Financing and investors 2	Funding for consortiums projects - international cooperation agencies	IDB, World Bank, Development Bank of Japan, Green Climate Fund, KfW, CAF, CIF
	Funding for consortiums projects – Institutional investors	Pension funds, wealth management, etc.

5.1 & 5.2/ Stakeholder groups (2/3)

Not exhaustive

Group	Role(s)	Organizations (XX Possible champions)
Technology development 3	Production of FCEVs and hydrogen technology	Hyunday motor company, Toyota, BMW; LIEBHERR; Alset, hydrogenics, Audi, Faurencia, Daimler
	Production of hydrogen driven mining truck vehicles	Ballard; Caterpillar; Komatsu
	Development of hydrogen aircrafts	Boeing, GE aviation, Airbus, Safran
	Development of ammonia ships & vessels	Maersk, Ultramar
Energy generation 4	Renewable energy generation	**Enel, Colbún, AES Gener; Engie,** Repsol, EDPR, Statkraft, RWE, Orsted
	Wind energy generation infrastructure	Vestas, Orsted, Nordex, Siemens, GE
	Solar energy generation infrastructure	JinkoSolar, SunPower, Hanwha Q Cells
	Infrastructure required for energy transmission	Transelec
Hydrogen production 5	Production of hydrogen technology	**Siemens, Cummins,** Nel Hydrogen, Hydrogenics, ITM Power
	Hydrogen production, transport and storage	**Total, COPEC group, Repsol, Engie,** Air liquid, BP, Nel Hydrogen, Shell, Vopak, ITM power
	Production of green ammonia and blue fertilizers	**Yara, Mosaic,** Air Liquid (Airgas), BASF
	Production of blue methanol	Methanex

5.1 & 5.2/ Stakeholder groups (3/3)

Not exhaustive

Group	Role(s)	Organizations
Ecosystem builder	Coordination and orchestration of different actors - public and private sector Supporting the design of the regulatory framework	Ministry of Energy
	Facilitating the establishment of international cooperation agreements	Ministry of Foreign Affairs; Embassies at target countries including: Japan, South Korea, China, EU and USA
	Development of funding lines for initial pilots and feasibility studies via CORFO and promoting the country as an investment destination	Ministry of Economy, Development and Tourism
	Development of infrastructure studies and coordination with private organizations for infrastructure development	Ministry of public works
	Promotion of the adoption of green ammonia and low-carbon fertilizers produced with hydrogen	Ministry of agriculture
	Establishment of environmental requirements and processes for the development of hydrogen projects	Ministry of environment
	Promoting and coordinating together with universities the research on hydrogen and derivatives	Ministry of Education, Ministry of Science, Technology, Knowledge and Innovation
	Design the permits for the operation of hydrogen pilots and coordinate the work with companies	Ministry of mining, Ministry of transportation
	Promotion of research and education on green hydrogen	U. de Chile, PUC, U. de Concepción, U. de Magallanes, U. de Antofagasta, UTFSM
	International research and development partners	Fraunhofer, Hydrogen Energy Research Center
6	Representation of the private sector and coordination with public entities	Hydrogen council, H2 Chile, Hydrogen Europe, Ammonia Energy Association

XX Possible champions

5.3/ Conditions which favor the participation of new actors and potential impacts

Conditions which favor new actor participation

Aspiration, roadmap and strategic priorities

Transparent aspiration and roadmap to show government support for green hydrogen industry development, defined strategic priorities to enable companies to identify and capitalize on strategically aligned opportunities

Regulation & licensing / permitting

Establishing stable regulatory frameworks which support long term investment and development, are easy to understand (particularly for international players) and have a clear point of contact to field questions and doubts

Effective coordination

Creating effective communication and coordination through the establishment of specific task forces (responsible for specific targets) and governance structures (responsible for the overall advancement of the roadmap)

Potential impact of new actor participation

Development of consortiums (supply & demand)

Consortium launch for domestic and international applications (and most likely by both domestic and international players), providing both supply, demand and financing requirements for hydrogen industry launch

Attraction of international financing

The establishment of consortiums (and in particular, the involvement of international actors) will facilitate the attraction of international financers (ex. national banks may become more interested if national companies are taking part in the consortium or if consortium actors have pre-existing relationships with international financers)

Reduced costs through synergies

Increased coordination and partnership building will create cost reducing synergies, particularly in infrastructure, which will accelerate green hydrogen industry development

5.4/ Workshop 1: Private Actors

Workshop for private actors

Objectives

Review Chile´s National Green Hydrogen Strategy and clarify questions

Discuss the principal opportunities, barriers and demand offtake sources for the establishment of successful projects in Chile

Prioritize these opportunities, barriers and demand offtake sources for further development by the Ministry of Energy

Gain visibility on the immediate next steps for the hydrogen industry in Chile

Agenda

1	**Introduction**	**10:00 – 10:05**
2	**Review Chile´s National Green Hydrogen Strategy**	**10:05 – 10:35**
3	**Q&A on national strategy**	**10:35 – 11:00**
4	**Small group breakout: building a consortium**	**11:00 – 11:40**
5	**What´s next?**	**11:40 – 12:00**

5.4/ Workshop 1: National Strategy for Hydrogen Breakout group discussion

Breakout group assignments

Consortium A: Ammonia export

- H2 Chile, Generadoras, Engie, AME, Mitsui, Sumitomo, GasValpo, RWE, Austrian Solar, Siemens, Air Products
- Facilitator: Xavier Costantini; Note taker: Daniela Hermosilla

Consortium B: Mining sector

- UTFSM, H2 Chile, Cummins, CAP, COPEC, Transelec, Pacific Hydro, TCI GECOMP, ENAEX
- Facilitator: Clemens Muller-Falcke; Note taker: María José Lambert

Consortium C: Domestic applications (natural gas, logistics, oil refinery)

- PUC, ACERA, Walmart, Metrogas, COPEC, Lipigas, GasValpo, ENAP, CCU
- Facilitator: Sarah Toupal; Note taker: Benjamín Maluenda

Discussion questions

- Do you believe in the **hydrogen opportunity** for your consortium (in what timeline?)
- What are the **barriers** that stand in the way? How can they be overcome?
- What is required to **secure demand offtake** for your consortium?

Workshop 2: Public Actors

Objectives

Introduce Chile´s National Green Hydrogen Strategy and clarify questions

Discuss how hydrogen can help support the goals of each government area and what challenges exist to the successful implementation of the strategy

Gain visibility on the immediate next steps for the hydrogen industry in Chile

Agenda

1	**Introduction**	**9:00 – 9:10**
2	**Review Chile´s National Green Hydrogen Strategy**	**9:10 – 9:40**
3	**Q&A on national strategy**	**9:40 – 10:20**
4	**Small group breakout: opportunities and barriers**	**10:20 – 10:50**
5	**What´s next?**	**10:50 – 11:00**

Breakout group discussion

Dynamic

Divide into small groups

Break out (25 mins)

- Discuss and answer questions, supported by facilitators

Break out wrap up (5 mins)

- Facilitators present results from each breakout group

Discussion questions

What **opportunities** does hydrogen offer to your area / institution (how can hydrogen **support your goals**)?

In thinking about your area, what are the **principal challenges** you see in launching the hydrogen industry in Chile?

Proposed template to fill with your teams: articulate the opportunities and challenges you see for your department across our eight domains for work

Thinking beyond the high level actions outlined in the national strategy…

Name: **Department:**

1 Strategy and targets

2 Regulation and licensing / permits

3 Coordination and partnerships

4 Value chain

5 Financing and incentives

6 Infrastructure

7 Research & Development

8 Skilled Labor

Chapter 6: government role in hydrogen industry development

Chapter content description

- **Domains of work** required to support and accelerate the development of the hydrogen industry in Chile (both demand and supply) and definition of key design principals
- **List of actions taken by international governments** to implement the domain of work according to the key design principal
- **Recommendation of immediate next steps** for government action by domain to accelerate and successful implement the hydrogen industry roadmap

Activities included

- **Activity 6.1**
- **Activity 6.2**
- **Activity 6.3**
- **Activity 6.4**

1. End use applications prioritized based on market potential and impact on strategic roadmap and do not represent an exhaustive list. Some end use applications such as ´energy storage´ were included in the original scope of work but deprioritized in favor of replacement of grey hydrogen with green hydrogen, an end use application with the potential to reach economic break even in the short term.

6.1-6.4/ The state will support the development of the industry taking action in 8 domains

Not exhaustive

Enabling environment

1 Strategy and targets

Establish a clear vision with targets, strategy, and action plan.

Communicate the strategy and play an active role in mobilizing private and public actors in Chile and beyond.

2 Regulation and licensing / permits

Establish a long term regulatory structure which diminishes uncertainty and minimizes barriers to entry / doing business

Accelerates and streamline licensing and permitting processes

3 Coordination and partnerships

Coordinate and establish formal partnerships with domestic and international actors to promote industry growth and domestic and international supply chain development

4 Value chain

Maximize value creation and economic development in Chile identifying where the opportunities are for domestic industry along the value chain and putting the right enablers in place

5 Financing and incentives

Provide sources of funding directly and facilitate access to inexpensive capital

Deployment incentives (production and demand)

6 Infrastructure

Promote infrastructure developments via PPPs and collaboration among actors to capture infrastructure synergies

7 Research & Development

Promote research projects on hydrogen technologies both nationally and with international partners

8 Skilled Labor

Develop local talent and skills required for H2 production, refueling, storage, etc.

Source: Press research; Hydrogen roadmap for EU, USA; Netherlands, France, Australia, Germany, South Korea; Team analysis

6.1-6.4/ Enabling actions (1/4)

Not exhaustive

Not exhaustive

Domain	Key design principles	Examples of actions seen in other places
1 **Vision & targets**	Set clear ambition and direction Create strategy, clear targets and an action plan to support ambition Actively mobilize public and private actors to realize defined strategy (in Chile an internationally) Communicate the benefits of the plan to the society in general	Define national aspiration backed by **strategy, targets and action plan**, including priorities for industry development and the role of the government and industry actors Mobilize public and private actors to **capture strategic opportunities** aligned with the aspiration (ex. developing off taker markets, facilitating international financing etc…) Understand the full **economic and social impact** of the Hydrogen strategy (both supply and demand side) Create the **socialization and engagement strategy** for the society in general
2 **Regulation**	Diminish the uncertainty to invest in new projects Facilitate market access Establish safety standards Lay down environmental requirements	Formulate the **regulatory framework and protocols** for safe production, storage and transportation of hydrogen Establish the **rules for grid integration** and remuneration for energy production and grid stabilization Create a **certification scheme or guarantee of origin** to recognize greener supply chains that use green hydrogen
Licenses / permits	Establish clear requirements Design transparent and fast processes Reduce bureaucracy	Establish clear and transparent rules and processes for **license granting** for H2 projects (plants, refueling stations, storage, etc.) Design a **streamlined and transparent application process** with clear steps and one only central point of contact Facilitate licenses to access to **water sources**

Source: Press research; Hydrogen roadmap for EU, USA; Netherlands, France, Australia, Germany, South Korea; Team analysis

6.1-6.4/ Enabling actions (2/4)

Not exhaustive

Not exhaustive

Domain	Key design principles	Examples of actions seen in other places
3 **Coordination with local actors**	Orchestrate the efforts of different actors Capture potential synergies between actions Balance H2 supply and demand growth	Create a **task force to lead the implementation of the strategy** and the orchestration with the different sectors Coordinate the **procurement activities of industry actors** to promote bulk purchases of different technologies (e.g. buses, HDT)
Partnerships with technology providers and investors	Establish partnerships with equipment providers to attract and incorporate new technologies	Establish **partnership with PV / Wind energy developers and electrolyzer producers** to acquire equipment through bulk / long term contracts, including the attraction of production or assembly lines in Chile Establish consortiums with national and **international ammonia and fertilizers producers** to attract investments (e.g. mosaic, yara) Establish **partnerships with technology producers** (e.g. hyundai, caterpillar) to support the technology development and attract early pilots to the country
Country to country partnerships	Facilitate market access for exports / imports	Establish **free trade agreements and country-to-country collaboration partnerships** to facilitate imports / exports of technology and products
4 **Value chain development**	Identify strategic opportunities for economic development in the value chain (upstream and downstream) Create mechanisms which support local value chain development	**Map green hydrogen value chain activities**, both upstream and downstream, prioritizing opportunities based on attractiveness (e.g. size and value-added) and fit with local capabilities and advantages **Develop a "Zona de franca"**: North & South to attract manufacturing investment and develop domestically value adding activities of the supply chain

Source: Press research; Hydrogen roadmap for EU, USA; Netherlands, France, Australia, Germany, South Korea; Team analysis

6.1-6.4/ Enabling actions (3/4)

Not exhaustive

Not exhaustive

Domain	Key design principles	Examples of actions seen in other places
5 **Financing**	Provide cheap financing for early proof-of-concepts and pilots to accelerate industry growth without creating industry dependence on government support Attract projects and investments with high scale-up potential and secure long term contracts (potential lock-ins) Protect private investor rights	Coordinate actors from the **financial sectors to establish investment lines** for the development of the hydrogen ecosystem Provide financial support for **initial feasibility studies** with commitment for further scale-up without government financing Create financing lines for the initial **construction of green ammonia and blue urea factories** Fund **pilot projects to test technologies in key applications** (mining, transport, H2 replacement, gas blending, etc.) with potential further development without government funding Establish **financing lines for fleet renovation** (e.g. buses, HDT) with FCEV and/or **for shared refueling station** at key hubs Advocate for regulation which **reduces risks for the deployment of capital** for green hydrogen projects in Chile
Incentives	Support initial investments in sectors with higher potential of long-term socio-economic gains Apply the most efficient tools to anticipate the industry development without creating industry dependence on government support	Permit hydrogen producers to use the **grid for transmission at marginal costs** (not full cost) Create **carbon bonds** for green energy producers and carbon capture in urea production **Create a Carbon tax** per ton of CO2 for transport applications **Establish royalties abatements** for mining companies to incentive investment in FCEV (haul trucks and HDT) – for a limited period of time **Import duty on grey ammonia and derivatives** Increase **subsidies for green fertilizers** in the market Establish **emissions standards for vehicles** and incentives for FCEVs

Source: Press research; Hydrogen roadmap for EU, USA; Netherlands, France, Australia, Germany, South Korea; Team analysis

6.1-6.4/ Enabling actions (4/4)

Not exhaustive

Not exhaustive

Domain	Key design principles	Examples of actions seen in other places
6 **Infrastructure**	Facilitate infrastructure development to enable hydrogen / derivative supply and demand Capture synergies in infrastructure development Leverage infrastructure already in place	Establish protocols and rules for **repurposing of existing infrastructure (ports, pipelines, etc.)** Develop **pipeline network** for hydrogen and ammonia transport (from production to utilization sites) Develop **port infrastructure and road transportation routes** via PPPs Build **transmission line** infrastructure at scale Facilitate the installation of **refueling stations** at key hubs (e.g. bus terminals, ports, etc.) Reinforce the **Infrastructure of key cities / regions** to make them more attractive for skilled people
7 **Research & development**	Enable rapid technology development, transfer and adoption Partner with other governments to conduct joint R&D on key topics for Chilean H2 development	Create a **research center for green hydrogen** that promotes and coordinate research efforts between among bodies and companies Create **lines of funding for research (e.g. Fondecyt) on hydrogen** related issues at universities and R&D centers of companies Develop **research projects together with international research centers / universities** on key issues for the Chilean hydrogen economy
8 **Skilled Labor**	Prepare sound technical experts for the industry Give job opportunities to disadvantage communities Promote gender equality	Promote **programs and courses on** hydrogen **related topics** in universities and technical centers Support **worker training and educational programs in** hydrogen production centers, refueling stations, storage points, among others

Source: Press research; Hydrogen roadmap for EU, USA; Netherlands, France, Australia, Germany, South Korea; Team analysis